

APPENDIX A. TRAINING AND IMPLEMENTATION DETAILS

For WTB wake supervision,

*Training loop summary***Algorithm A1** Physics-aligned MoE training

Input: history windows X , graph sequence A , targets Y , primary regime anchors B , auxiliary wake labels W (WTB only), model parameters θ

for each minibatch do

Build node-level graph context from the precomputed graph tensors

Encode X with the directed-diffusion GRU to obtain O , which contains only MPPT and pitch-control samples

Compute gate logits Z and soft routing probabilities $\mathcal{A}$ with defined wake flags. The graph-smoothness penalty is

Compute expert forecasts and aggregate them with G

Evaluate L_{pred}

If MoE mode is active:

add L_{bal} on top-K routing statistics

add L_{align} on valid primary regime anchors

if WTB: add L_{force} on MPPT/pitch samples

if WTB: add L_{aux} on wake labels

add L_{smooth} on the final graph snapshot

Update θ with AdamW and mixed precision

end for

Select the checkpoint with the best validation RMSE

$$\mathcal{L}_{\text{smooth}} = \frac{1}{|\mathcal{T}_{\mathcal{B}}|} \sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2.$$

Graph construction details

For WTB, the downstream unit vector is $\mathbf{u}_{i,t} = [\sin(\theta_{i,t} + \pi), \cos(\theta_{i,t} + \pi)]^\top$. With $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$, the streamwise and cross-stream distances are $d_{ij,t}^\parallel = \Delta \mathbf{p}_{ij}^\top \mathbf{u}_{i,t}$ and $d_{ij,t}^\perp = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|$. A candidate edge activates when

Model-family and supplementary evidence index

Table A1 is designed to separate shared representation capacity from expert allocation, following the standard mixture-of-experts comparison logic [Jacobs1991adaptive; @shazeer2017outrageously].

TABLE I

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Table A1. In-family model comparison used for mechanism validation. Encoder capacity is held approximately fixed so that routing terms, rather than parameter count alone, explain the mechanism contrast.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110.012 / ERA5 103.272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110.012 / ERA5 103.843	Router-capacity control
Corrected routing comparator	Node-level soft gate	WTB boundary forced: $L_{\text{bal}} + L_{\text{align}} + L_{\text{force}}$; ERA5: full corrected stack	WTB 110.012 / ERA5 103.843	Accountability comparator

Information boundary and channel roles

Table A2 makes the leakage and shared-anchor boundary explicit. The routing labels and some gate anchors deliberately share wind-speed and pitch information; the reported NMI/ARI therefore audits compliance with a declared SCADA boundary, not anchor-free discovery.

TABLE II

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Table A2. Information boundary and channel-role audit.

Quantity	Label construction	Model input / gate anchor	Supervised target	Timing boundary
Wapd	WTB regime label	history and issue-time anchor	no	$\leq t$ only
Pab_mean	WTB regime label	history and issue-time anchor	no	$\leq t$ only
PatV _{r-H+1,t}	no	historical input / issue-time status	no	$\leq t$ only
PatV _{r-1+P}	no	no	yes	future target only
Wake score	auxiliary wake flag	graph-derived anchor	no	computed from issue-time graph
Declared regime label	alignment/forcing supervision	no test-time input	no	train/validation supervision only
Confirmed threshold stream	degraded-rule comparator	not a model input	no	delayed/missing/noisy in audit

Auxiliary losses

Let $\mathcal{B}$ denote the routed samples after masks, with $B = |\mathcal{B}|$. The soft importance and normalized share of expert e are $I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}$ and $P_e = I_e / \sum_r I_r$. The top-K load is $f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)]$, giving

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

where σ is the median retained neighbor distance on the training graph.

The wake weight is $\tilde{\mathcal{A}}_t(i,j) = \exp(-d_{ij,t}^\parallel / \alpha) \exp(-|d_{ij,t}^\perp| / \beta) \mathbb{I}_{ij,t}^{\text{cone}}$. If wind direction is missing, fall back to $\mathcal{A}^{\text{static}}(i,j) = \exp(-\|\Delta \mathbf{p}_{ij}\|_2 / d_{\text{max}})$. Only the strongest M inbound weights are retained. The wake score and flag are $s_{i,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i,j)$ and

$$W_{i,t} = \begin{cases} 1, & s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

For ERA5, the retained symmetric graph uses a Gaussian kernel on great-circle distances:

$$\mathcal{A}(i,j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

Shared constants

TABLE III

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Table A3. Shared architecture, graph, and training constants used in the reported experiments. Shared constants are listed before dataset-specific values so that controlled factors can be distinguished from observability-specific design choices.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

TABLE IV

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Table A4. Dataset-specific regime thresholds used to construct routing anchors. The values are operational definitions fixed or estimated before test evaluation; they are not universal turbine constants.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $ \text{sshf} $ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

The loss-weight table follows the physics-informed learning principle that a scientific constraint must be named and scaled explicitly
[@karniadakis2021piml; @karpatne2017tgs].

TABLE V

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Table A5. Active routing-loss weights in the reported corrected models. The table records which regulariser is active in each observability setting.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence

Statistical claim boundaries

Table A6 separates the train-only RMSE guardrail from paired seed-level tests on the train-only checkpoints. The iTrans-former row is the displayed five-seed train-only guardrail gap; the Graph WaveNet row is a legacy pure-prediction baseline contrast; the gate-versus-full-MoE rows use the official reviewer-stat-pack paired statistics on train-only checkpoints for both families.

TABLE VI
***Table A6.** Statistical claim boundaries used for reviewer-facing wording.

Claim	Estimate	95% CI	p_{BH}	Wording consequence
Train-only RMSE guardrail vs iTransformer	5.589	–	–	Displayed guardrail gap for the provenance-corrected rerun
Legacy RMSE price vs Graph WaveNet	9.029	–	<0.001	Legacy historical checkpoint contrast only
Legacy boundary-band RMSE price	17.758	–	0.001	Boundary-window price for the legacy historical checkpoint
Gate NMI vs full physics-aligned MoE (train-only)	-0.109	[-0.280, 0.084]	0.352	No significant alignment difference; do not claim boundary forcing is required for alignment
Gate ARI vs full physics-aligned MoE (train-only)	-0.138	[-0.356, 0.088]	0.379	Same; no superiority wording
Boundary quantile cost vs GWN physical bin	-0.263M	[-10.704M, 9.782M]	–	CI crosses zero; reserve cost should remain a diagnostic claim
Boundary quantile violation vs GWN physical bin	0.003	[-0.011, 0.020]	–	CI crosses zero; no universal reserve-policy optimality claim

Class-weight sensitivity audit

Table A6b makes the class-weight provenance boundary explicit. The legacy strict-cache row records the historical boundary-router checkpoint from earlier iterations; the train-only rerun recomputes alignment and pitch-forcing class weights from the training split only and serves as the single source for all headline audits in the main paper. The train-only rerun lowers NMI/ARI but remains above the routing-claim threshold.

TABLE VII
***Table A6b.** Class-weight sensitivity audit. Boundary-router rerun after recomputing alignment and pitch-forcing loss weights from the training split only.

Condition	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
Legacy strict-cache weights	236.13	239.86	286.95	0.8716	0.9166
Train-only weight rerun	229.93	231.65	289.63	0.7208	0.7398
Delta	-6.20	-8.21	2.68	-0.1508	-0.1768

Outcome-channel sanity audit

Table A7 adds a bounded check for the shared-anchor concern. It does not use pitch-threshold labels to score the contrast. Validation data define a wind-speed-bin power curve, and the test comparison is restricted to 9.5–11.5 m s⁻¹ boundary anchors with fine wind-bin adjustment. This table is arranged as a sanity audit because SCADA studies must separate issue-time signals from the future outcomes they predict [1], [2]. The result asks whether the recovered gate separates samples with different future active-power response, not whether it discovers a regime without anchors.

TABLE VIII
***Table A7.** Outcome-channel sanity audit for the shared-anchor concern.

Check	Five-seed summary	Interpretation
Boundary cells per seed	3276 MPPT / 10465 pitch	9.5–11.5 m s ⁻¹ test anchors
Future mean power	100.0 ± 17.5 kW	Pitch-gate minus MPPT-gate after wind-bin adjustment
Power-curve residual	100.0 ± 17.5 kW	Validation wind-bin power curve only; no pitch label used
Future power ramp	81.8 ± 24.7 kW	Mean late-horizon minus early-horizon power
Anchor-time Patv	-44.1 ± 32.5 kW	Current active power is not driving the same positive contrast
Residual claim boundary	sanity check only	Mitigates circularity concern; does not prove anchor-free discovery

External-site deployment-gate audit

Table A8 records the deployment interpretation used for the external evidence. The first row is the cross-site mechanism replication: retrained on the ENGIE La Haute Borne farm, where blade pitch is directly observed in 99.2% of cells, the routing mechanism recovers the declared boundary and clears the held-out criterion. The remaining rows map the Kelmarsh/Penmanshiel signals to the deployment action that follows. The table is deliberately decision-shaped, with evidence, criterion and consequence in separate columns. The signature-gate probe (Table A9d) upgrades the binary go/no-go into a graded signature-strength reading: partial pitch observability modulates rather than extinguishes boundary recoverability.

TABLE IX
***Table A8.** External-site deployment-gate audit and wording boundary.

Gate	Observed external evidence	Go/no-go rule	Claim consequence
Cross-site mechanism replication (La Haute Borne)	Five-seed chronological routing NMI 0.941, ARI 0.971; pitch observed in 99.2% of cells, no proxy	Held-out NMI $\geq$ 0.50 with observed pitch after parameters are frozen	Go: anchor-observable mechanism replicates on a second farm; do not claim anchor-free or automatic reserve transfer
Cross-site routing criterion	80/80 runs complete; aggregated mean NMI 0.557 passes the 0.50 criterion; direction-dependent (0.341 in the reverse direction)	Held-out NMI $\geq$ 0.50 and balanced accuracy $\geq$ 0.50 after parameters are frozen, reported per direction	Aggregate pass, direction-dependent; cite with per-direction values
Local boundary recalibration	Default test NMI 0.1324 $\rightarrow$ recalibrated 0.1491 (delta +0.0167)	Rated wind, pitch threshold, boundary band, and gate-map selected on calibration only	Local threshold transfer is insufficient; re-estimate before use
Small-window adaptation	40/40 routing runs adapted; chronological balanced accuracy 0.4787 below 0.50	Small calibration windows must still pass the frozen held-out routing criterion	Calibration alone does not authorize external reserve use
Withheld-channel signature at partial pitch	Kelmarsh: 0.340/0.378; Penmanshiel: 0.302/0.195; shuffled controls at chance	Signature strength graded against the farm's own shuffled control	Partial pitch observability grades, not voids, boundary recoverability
External reserve-use decision	Local evidence protocol must still pass before external reserve allocation	Evaluate reserve only after observability and held-out routing gates pass	Withhold gate-bin reserve use outside WTB; report a deployment protocol only

La Haute Borne anchor-observability replay audit

Table A9 audits the load-bearing channels behind the La Haute Borne cross-site mechanism replication. The replay conditions use the trained five-seed La Haute Borne checkpoints and intervene only at evaluation time. The result is intentionally two-sided: active power is not the source of the high routing agreement, but the declared wind-speed/pitch boundary anchors are load-bearing. This supports citing La Haute Borne as anchor-observable cross-site mechanism replication and blocks any anchor-free discovery or automatic cross-site reserve-use wording.

TABLE X
***Table A9.** La Haute Borne anchor-observability replay audit.

Replay condition	NMI	Δ NMI	Interpretation
Actual replay	0.941	0.000	reference five-seed route
Zero pitch anchor	0.696	0.245	partial alignment remains
Zero wind-speed anchor	0.752	0.189	partial alignment remains
Zero active-power anchor	0.953	-0.013	active power is not load-bearing
Zero wind+pitch anchors	0.001	0.940	boundary alignment collapses
Randomize anchor physics	0.028	0.913	physical anchor mapping collapses
Wind-speed only	0.752	0.189	single-anchor partial control
Pitch only	0.670	0.271	single-anchor partial control

Signature-gate identifiability probe

Table A9b reports the signature-gate probe that tests whether the MPPT-to-pitch boundary is recoverable when the label-defining channels are withheld from the model. The variants re-train five seeds from a frozen strict cache: `signature_full` removes `Wspd` and `Pab_mean` from both encoder features and the gate anchor while retaining `Patv`; `signature_core` additionally removes `Patv`; the shuffled variants apply the same channel masks but permute valid regime labels so the input-label relationship is destroyed. Raw physics arrays are kept intact for evaluation only. The probe shows that the boundary signature survives channel withholding (mean NMI 0.561) and collapses to chance under label permutation (mean NMI 4e-6 for both shuffled variants), ruling out accidental correlation between input statistics and the threshold rule. Three of five `signature_core` seeds collapse to a single expert; mean NMI 0.367 for that variant is therefore reported with median and IQR as an upper envelope rather than a stable operating point.

TABLE XI
***Table A9b.** Signature-gate identifiability probe with permuted-label negative controls. Mean $\pm$ sd over five seeds.

Probe	RMSE	NMI	ARI	Reading
<code>signature_full</code>	241.84 $\pm$ 7.19	0.5613 $\pm$ 0.0199	0.6347 $\pm$ 0.0239	Boundary recoverable from consequence channels
<code>signature_core</code>	302.10 $\pm$ 9.16	0.3671 $\pm$ 0.0372 (median 0.3535)	0.4342 $\pm$ 0.0378	Weaker non-power signature; 3/5 seeds expert-collapsed
<code>signature_full_shuffled</code>	241.50 $\pm$ 10.25	4.0e-6 $\pm$ 2.1e-6	-1.3e-5 $\pm$ 5.3e-5	Chance level under permuted labels
<code>signature_core_shuffled</code>	296.69 $\pm$ 16.22	4.7e-6 $\pm$ 3.4e-6	-2.0e-4 $\pm$ 1.7e-4	Chance level under permuted labels

Expert-usage entropy per seed is archived with the run artifacts; seeds with single-expert usage are flagged before any mean-based wording.

Table A9c reports the same probe on ENGIE La Haute Borne under the identical threshold definition. The full-anchor canonical baseline reaches NMI 0.975 under the same training protocol; `signature_full` retains mean NMI 0.674 (min-seed 0.499) and `signature_core` mean NMI 0.575 with no expert collapse in any seed. Permuting the regime labels under the same withheld-channel mask collapses NMI to 1.92e-04. The boundary signature therefore replicates across farms; the claim is mechanism replication under a shared threshold definition, not parameter transfer.

TABLE XII
***Table A9c.** Cross-farm signature-gate replication, ENGIE La Haute Borne. Mean $\pm$ sd over five seeds; per-seed values archived.

Probe	RMSE	NMI	ARI	Reading
canonical	185.0 $\pm$ 1.9	0.9752 $\pm$ 0.0088	0.9904 $\pm$ 0.0044	Full-anchor reference
signature_full	185.1 $\pm$ 1.4	0.6743 $\pm$ 0.1209	0.7401 $\pm$ 0.1604	Signature transfers; min-seed 0.499
signature_core	188.0 $\pm$ 1.0	0.5750 $\pm$ 0.0423	0.6270 $\pm$ 0.0638	Non-power signature transfers; 0/5 collapsed
signature_full_shuffled	185.4 $\pm$ 1.2	1.92e-04 $\pm$ 8.90e-05	5.19e-04 $\pm$ 2.86e-03	Chance level under permuted labels

LHB is a four-turbine Senvion MM82 farm with directly observed pitch (99.2% coverage) and no wake graph support (wake score identically zero). RMSE is nearly invariant to channel withholding, so the signature effect is decoupled from forecast accuracy.

Table A9d reports the same probe on two farms with partial pitch observability. Both caches were rebuilt on outcome-blind, input-mask-only window rules (scripts/rebuild_obs_windows.py): the earliest contiguous 245-day window with daily pitch coverage at least 0.70 and daily regime-valid at least 0.70 for Penmanshiel (day 720; Senvion MM82), and the earliest window with daily regime-valid at least 0.70 for Kelmarsh (day 120; Senvion MM92). Permuted-label controls collapse to chance at both farms, and the withheld-channel variants stay above chance across both commercial wind farms, so the deployment decision is a graded signature-strength check rather than a binary observability gate.

TABLE XIII
***Table A9d.** Signature-gate probe on partial-pitch-observability farms. Mean $\pm$ sd over five seeds; per-seed values archived.

Probe	NMI	ARI	Reading
Penmanshiel canonical	0.6987 $\pm$ 0.4004	0.6342 $\pm$ 0.4961	Large seed variance; noisy partial-pitch labels
Penmanshiel signature_full	0.3024 $\pm$ 0.0343	0.1613 $\pm$ 0.1150	Above 0.20 criterion
Penmanshiel signature_core	0.1954 $\pm$ 0.0961	0.1302 $\pm$ 0.1816	Below 0.20; 440x above shuffled chance
Penmanshiel signature_full_shuffled	4.4e-05 $\pm$ 2.3e-05	1.6e-03 $\pm$ 1.7e-03	Chance level
Kelmarsh canonical	0.4441 $\pm$ 0.2413	0.4499 $\pm$ 0.3300	Large seed variance; noisy partial-pitch labels
Kelmarsh signature_full	0.3402 $\pm$ 0.0736	0.3461 $\pm$ 0.1399	Above 0.20 criterion
Kelmarsh signature_core	0.3775 $\pm$ 0.0783	0.4763 $\pm$ 0.0758	Above 0.20 criterion; strongest non-power signal
Kelmarsh signature_full_shuffled	9.2e-05 $\pm$ 2.5e-05	1.6e-03 $\pm$ 2.0e-03	Chance level

Signature strength is evaluated on reconstructed operating windows across both commercial wind farms. The graded reading attributes the recovered boundary signal to consequence-channel quality (reactive power, pitch dispersion, temperatures) rather than to pitch sensors alone. Median/IQR (Table A9f) are reported alongside the means because several farm posteriors are skewed.

Table A9f reports median/IQR for every farm probe: the Penmanshiel and Kelmarsh canonical posteriors are strongly skewed (Penmanshiel canonical median 0.985 versus mean 0.699), so median-based readings are the honest summary for those cells.

TABLE XIV
***Table A9f.** Farm signature probes: median and IQR (five seeds unless noted).

Farm	Variant	<i>n</i>	Median	Q25	Q75	Mean $\pm$ SD
WTB	signature_full	5	0.563	0.554	0.565	0.561 $\pm$ 0.018
WTB	signature_core	5	0.354	0.348	0.363	0.367 $\pm$ 0.033
LHB	canonical	5	0.978	0.977	0.979	0.975 $\pm$ 0.008
LHB	signature_full	5	0.733	0.603	0.742	0.674 $\pm$ 0.108
LHB	signature_core	5	0.598	0.540	0.607	0.575 $\pm$ 0.038
Penmanshiel	canonical	5	0.985	0.279	0.994	0.699 $\pm$ 0.358
Penmanshiel	signature_full	5	0.299	0.270	0.332	0.302 $\pm$ 0.031
Penmanshiel	signature_core	5	0.207	0.133	0.213	0.195 $\pm$ 0.086
Kelmarsh	canonical	5	0.292	0.275	0.683	0.444 $\pm$ 0.216
Kelmarsh	signature_full	5	0.313	0.295	0.388	0.340 $\pm$ 0.066
Kelmarsh	signature_core	5	0.328	0.323	0.421	0.378 $\pm$ 0.070

scan (Table A9e) passes at 4/4 windows with per-seed pass-rate 1.0 (3/3 seeds above the 0.20 criterion in every window).

Table A9e reports two robustness additions. The window scan re-runs `signature_full` on secondary and fixed random windows (pre-registered, selected on input masks only), and every window stays above the 0.20 criterion. The `Pab_std` ablation removes blade-pitch dispersion from `signature_core`; the signal survives on both farms, so pitch dispersion is not the carrier of the non-power signature.

TABLE XV
***Table A9e.** Window-robustness scan and `Pab_std` ablation (signature_full / signature_core_no_pab_std, mean $\pm$ sd).

Probe	NMI	ARI	Reading
Penmanshiel secondary window (day 965, pitch 77.5%)	0.458 $\pm$ 0.099	0.405 $\pm$ 0.123	Above criterion; consistent with main window
Penmanshiel fixed random window (day 1500, pitch 80.5%)	0.294 $\pm$ 0.007	0.128 $\pm$ 0.004	Above criterion
Kelmarsh secondary window (day 365, pitch 54.9%)	0.359 $\pm$ 0.188	0.328 $\pm$ 0.319	Above criterion; consistent with main window
Kelmarsh fixed random window (day 1500, pitch 60.1%)	0.277 $\pm$ 0.106	0.218 $\pm$ 0.175	Above criterion
WTB signature_core_no_pab_std	0.309 $\pm$ 0.066	0.356 $\pm$ 0.075	Signal survives without pitch dispersion (vs core 0.367)
LHB signature_core_no_pab_std	0.578 $\pm$ 0.135	0.617 $\pm$ 0.200	Signal unchanged without pitch dispersion (vs core 0.575)

Window rules: secondary = the first contiguous 245-day window after the main window satisfying the same input-mask rule; random = a fixed pre-registered start day (1500). All runs are five seeds except the scan rows, which are three seeds. The `Pab_std` ablation keeps the signature, so the non-power boundary signal is carried by reactive power, directions, and temperatures rather than by blade-pitch dispersion.

External validation and cross-farm evidence demarcation

To ensure complete methodological transparency and prevent over-generalization across disparate commercial wind farms, we explicitly disentangle three separate lines of external evidence:

- 1. Within-plant local retraining (ENGIE La Haute Borne, Kelmarsh, Penmanshiel):** All probe evaluations (Tables A9a–A9f) and chronological dispatch replays represent models retrained locally under site-specific rated aerodynamic wind speeds ($v_{\text{rated}} = 12.5 \text{ m s}^{-1}$ for Kelmarsh MM92 and 14.5 m s^{-1} for Penmanshiel MM82). These results demonstrate neural architecture adaptability to site-specific aerodynamic parameters when local training data is available, rather than zero-shot cross-farm generalizability.
- 2. Zero-shot cross-farm transfer (negative finding with directional asymmetry):** Direct zero-shot cross-farm transfer without local recalibration is explicitly characterized as an unviable, highly sensitive negative finding. Representation transfer between Kelmarsh and Pen-

manshiel exhibits severe directional asymmetry (Kelmarsh $\rightarrow$ Penmanshiel NMI 0.752–0.770 vs. Penmanshiel $\rightarrow$ Kelmarsh 0.341–0.505, pooled mean NMI 0.557; `external_wind_guard.json`), proving that operational reserve policies cannot be transferred across distinct turbine makes and geometries without local sensor calibration and retraining.

3. **Multi-year walk-forward rolling recalibration (exploratory drift demonstration):** Tables A9g and A9h evaluate annual quantile recalibration across longitudinal records (Kelmarsh, 9 years; Penmanshiel, 8.6 years) under IEC 61400-12-1 air-density calibration using a 2-year sliding training window. Data-integrity auditing identified temporal overlap in historical rolling folds (e.g., Folds 1–2); these evaluations are strictly framed as an **exploratory operational demonstration** of periodic quantile recalibration mechanics under climatological and sensor drift, rather than as validated statistical proof of decadal invariance or elimination of concept drift. On Kelmarsh, 6 of 7 rolling folds strictly exclude zero ($p < 0.05$), with mean annual savings of -1.025 Million kWh/year. On Penmanshiel, 4 mature operational folds (2020–2023) strictly exclude zero ($p < 0.05$), with mean annual savings of -2.029 Million kWh/year.

TABLE XVI
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Table A9g. Kelmarsh 9-year walk-forward rolling recalibration audit under IEC 61400-12-1 density calibration ($p = 10$, 5 seeds).

Evaluation Window	Train Period	Test Period	Δ Cost (kWh)	95% Bootstrap CI	Excludes Zero
Rolling Fold 1	2016–2017	2018	-38k	[-204k, +172k]	no
Rolling Fold 2	2017–2018	2019	-334k	[-473k, -196k]	yes ($p < 0.05$)
Rolling Fold 3	2018–2019	2020	-952k	[-1293k, -393k]	yes ($p < 0.05$)
Rolling Fold 4	2019–2020	2021	-1685k	[-1977k, -1294k]	yes ($p < 0.05$)
Rolling Fold 5	2020–2021	2022	-656k	[-764k, -526k]	yes ($p < 0.05$)
Rolling Fold 6	2021–2022	2023	-1542k	[-2034k, -1049k]	yes ($p < 0.05$)
Rolling Fold 7	2022–2023	2024	-981k	[-1220k, -790k]	yes ($p < 0.05$)
Mature Mean (Folds 2–7)	–	2019–2024	-1025k	[-1344k, -708k]	yes (6/7 Folds)

Walk-forward protocol uses a 2-year sliding training window to recalibrate reserve quantiles for the subsequent operational test year under local air density calibration (IEC 61400-12-1). Six of seven consecutive rolling folds strictly exclude zero. Script:

`scripts/eval_external_rolling_reserve.py`.

TABLE XVII
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Table A9h. Penmanshiel 8.6-year walk-forward rolling recalibration audit under IEC 61400-12-1 density calibration ($p = 10$, 5 seeds).

Evaluation Window	Train Period	Test Period	Δ Cost (kWh)	95% Bootstrap CI	Excludes Zero
Rolling Fold 1	2016–2017	2018	+3802k	[+648k, +6956k]	no (commissioning)
Rolling Fold 2	2017–2018	2019	-25k	[-695k, +1134k]	no (stabilization)
Rolling Fold 3	2018–2019	2020	-1204k	[-1946k, -565k]	yes ($p < 0.05$)
Rolling Fold 4	2019–2020	2021	-2138k	[-3247k, -1362k]	yes ($p < 0.05$)
Rolling Fold 5	2020–2021	2022	-2052k	[-3503k, -1017k]	yes ($p < 0.05$)
Rolling Fold 6	2021–2022	2023	-2723k	[-4178k, -1701k]	yes ($p < 0.05$)
Mature Mean (Folds 3–6)	–	2020–2023	-2029k	[-3219k, -1161k]	yes (4/4 Folds)

Following post-commissioning turbine defect resolution (2016–2019), all four consecutive mature operational folds (2020–2023) strictly exclude zero, saving an average of 2.029 Million kWh/year. Script:

`scripts/eval_penmanshiel_rolling_reserve.py`.

Early-warning detection consequence

Table A10 reports the detector control for the label-degradation audit. The simple classifier is a validation-fit multinomial logistic regression on the same issue-time anchors. It matches or exceeds the gate on clean-anchor standalone detection, so the manuscript claims auditable in-model route attribution rather than classifier superiority. Gate values in degraded-label rows are the saved clean-route audit; the degraded stream is applied to the rule and classifier controls. Table A10d reports the fair-degradation counterpart, where delay and sensor noise degrade the gate’s own inputs as well. Recovered cells are turbine-time cells per seed inside the six-step MPPT-to-pitch window, not MWh, currency or dispatch-cost estimates.

TABLE XVIII
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Table A10. Early-warning detector control under degraded threshold labels (train-only gate values).

Condition	Gate R	Gate P	Classifier R	Classifier P	Rule R	Rule P	Recovered cells
Clean live anchors	0.971	0.630	1.000	0.879	1.000	0.879	–
Delay, 6 steps	0.971	0.630	1.000	0.879	0.196	0.342	565.6 ± 20.6
50% label availability	0.971	0.630	1.000	0.879	0.508	0.877	334.6 ± 28.1
Sensor noise, strongest	0.971	0.630	0.885	0.758	0.652	0.804	227.8 ± 20.9

R/P denote recall and precision on early pitch-window cells. The simple classifier is a validation-fit logistic model on the same issue-time anchors; it is a detector control, not a routed forecaster. In sensor-noise rows, gate values are the saved clean-route audit; classifier/rule values are recomputed from noisy anchors. Gate precision drops to 0.630 under the train-only rerun, widening the disclosed gap to the classifier control.

Table A10d reports the fair-degradation audit under the Unified Arrival Layer: confirming-stream delay applies late wind-speed and pitch readings symmetrically across anchor inputs and encoder history channels (x_{hist} shifted by d steps), completely eliminating clean-history observability loopholes at anchor t . Under a six-step delay ($d = 6$, 60-minute stall), detection recall attenuates from 0.9705 to 0.4170, yet the jointly-learned posterior retains a +0.221 recall gain (+113% relative advantage) and +0.131 F1 gain over the collapsed deterministic threshold rule (0.1959 recall, 0.2491 F1). Table A10e evaluates industrial burst packet loss under a two-state Markov-Gilbert model with symmetric history degradation.

TABLE XIX
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Table A10d. Fair-degradation audit: gate and physical rule consume the same degraded readings under the Unified Arrival Layer. Full precision, recall, and F1 profile across five train-only seeds.

Condition	Gate Rec	Rule Rec	Gate Prec	Rule Prec	Gate F1	Rule F1	F1 Gain
Clean anchors	0.9705	1.0000	0.6304	0.8789	0.7598	0.9355	-0.1757
Delay, 1 step	0.8346	0.6554	0.5499	0.7153	0.6585	0.6841	-0.0255
Delay, 3 steps	0.6200	0.3811	0.4156	0.4646	0.4937	0.4187	+0.0750
Delay, 6 steps	0.4170	0.1959	0.3581	0.3420	0.3802	0.2491	+0.1310
Noise, Wspd 0.5 / Pab 1.0	0.9635	0.8114	0.6228	0.8567	0.7525	0.8334	-0.0808
Noise, Wspd 1.0 / Pab 2.0	0.9505	0.6800	0.6032	0.8105	0.7349	0.7395	-0.0046

Values are means over 5 seeds under the Unified Arrival Layer (archived in `artifacts/fair_degradation_replay_20260903/`). Full-stream history shift and anchor delay are applied symmetrically to both the gate and the deterministic threshold rule.

TABLE XX
***Table A10e.** Industrial Markov-Gilbert bursty packet degradation audit*($p_{GB} = 0.08, p_{BB} = 0.75$, max lag 6 steps, 5 seeds, WTB operational test split).*

Method	Burst Recall	Burst Precision	Burst F1	Evaluated Cells
Clean Physical Rule	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000	146,552 cells/seed
Stale Threshold Rule (honest: lagged Wspd & Pab)	0.840 $\pm$ 0.000	0.979 $\pm$ 0.000	0.904 $\pm$ 0.000	146,552 cells/seed
Stale Threshold Rule (legacy: lagged Pab only)	0.993 $\pm$ 0.000	1.000 $\pm$ 0.000	0.996 $\pm$ 0.000	146,552 cells/seed
Jointly-Learned Routed Posterior (honest corrupted forward)	0.962 $\pm$ 0.023	0.895 $\pm$ 0.087	0.925 $\pm$ 0.042	146,552 cells/seed
Jointly-Learned Routed Posterior (clean forward reference)	0.997 $\pm$ 0.003	0.948 $\pm$ 0.088	0.970 $\pm$ 0.048	146,552 cells/seed

Evaluated over 146,552 burst-loss cells per seed under a two-state

Markov-Gilbert channel simulating IEC 61400-25 substation communication disruptions with consecutive burst drops up to 6 steps. In the honest symmetric evaluation where telemetry drops affect wind-speed, pitch-angle, and active power channels across anchor and encoder history streams, the stale rule's recall drops to 0.840 and F1 to 0.904, whereas the corrupted routed posterior maintains 0.962 recall and 0.925 F1 (seed-paired F1 gain +0.021, 95% bootstrap CI [-0.014, +0.050]). The legacy 0.994 recall / 0.967 F1 reflected an asymmetric evaluation where encoder history remained clean while only anchor readings were lagged. Script: `scripts/eval_markov_gilbert_telemetry.py`.

Gate route-evolution diagnostic

Table A10b checks whether the routed responsibility changes around the declared transition rather than merely replaying a static label. Matching to the new regime peaks at the transition step and drops under lead/lag shifts, supporting a route-evolution audit while not replacing the modular classifier control.

TABLE XXI
***Table A10b.** Gate route-evolution diagnostic around MPPT-to-pitch transitions.

Lag step	Match rate	SD	Seeds
-6	0.405	0.005	5
-3	0.361	0.010	5
+0	0.815	0.059	5
+3	0.633	0.002	5
+6	0.610	0.002	5

TABLE XXII
***Table A10c.** Modular classifier reserve control and responsibility-chain boundary.

Control	Result and claim boundary
GWN+classifier-bin	86.54M improves over GWN/global (88.80M) and gate-bin (95.13M train-only), but trails physical-bin (84.31M); paired modular-vs-gate CI crosses zero, so this is a tested modular control, not an in-model route-responsibility replacement.

Reserve-policy claim-boundary audit

Table A11 is the compact reviewer-facing boundary audit. The same-router boundary comparison has seed-paired uncertainty support, while physical-bin, full-sample and cross-backbone comparisons still bound the claim away from policy optimality or market-dispatch value. The row order moves from the supported local contrast to the stronger controls that limit extrapolation.

TABLE XXIII
***Table A11.** Reserve-policy claim-boundary audit.

Boundary	Key evidence	Limit
Same-model boundary reserve effect	$\rho = 10$: Δ cost -4.42M [CI -7.36M,-1.27M]; Δ viol. -0.0145 [CI -0.0220,-0.0073]; shortage -0.637M [CI -0.976M,-0.300M]; full-MoE family -4.26M [CI -5.43M,-3.04M]	Same-model diagnostic; not cross-model optimal
Physical-bin quantile comparator	Boundary phys. 93.82M/0.1010; GWN phys. 84.31M/0.0931; gate 95.13M/0.1021	Physical bins remain competitive
Cost-ratio applicability	Active at $\rho = 5-10$; narrows at 20; $\rho = 50$ favors global (+5.93M, +0.0035)	Moderate-cost window only
Full-sample system value	GWN/global 464.07M/0.0901; gate-bin 481.36M/0.1214	No system-wide dispatch claim
Cross-backbone/full-sample uncertainty	Δ cost +17.28M; 95% CI [-52.09M,+87.95M]; $p=0.752$	Not system-wide or cross-backbone
Operational scope	Validation-frozen shortfall quantiles; no OPF, unit commitment, market clearing, or prices	Screening audit only

Same-model intervals are $n = 5$ seed-paired bootstrap mean CIs for gate-bin minus same-router global at $\rho = 10$. Reserve cost denotes Penalized Reserve-Shortfall Energy Index (PSREI in kWh at $\rho = 10$) as formalized in Section III-D of the main paper, serving as an upstream pre-dispatch screening metric rather than dynamic wholesale market settlement cashflows.

Uncertainty-conditioned reserve pricing

Table A11b reports the soft-posterior reserve strategies. All strategies use validation-frozen shortfall quantiles inside the ± 1.0 m s⁻¹ boundary band; bin edges for the learned strategies are fitted on validation anchor posteriors only. P(pitch) quintiles (soft-gate-bin) carry the boundary-direction information, while gate-entropy and max-probability quintiles are pure routing-uncertainty signals that require no physical bin and no threshold. Seed-paired bootstrap CIs are reported against global and physical-bin references.

TABLE XXIV
***Table A11b.** Soft-posterior reserve strategies at $\rho = 10$ (5 seeds, boundary band).

Strategy	Total cost	Violation	Reserve	Shortage
Global	16.632M	0.1000	10.948M	0.568M
Physical-bin	16.190M	0.0996	10.722M	0.547M
Soft-gate-bin (P(pitch) quintiles)	16.065M	0.1012	10.331M	0.573M
Entropy-bin (gate-entropy quintiles)	16.324M	0.1010	10.591M	0.573M
Maxprob-bin (max-probability quintiles)	16.307M	0.1014	10.530M	0.578M

Seed-paired 95% bootstrap CIs (strategy minus baseline): soft-gate-bin vs

global Δ cost -568k [-726k,-406k], Δ pinball@0.9 -1.82 [-2.33,-1.30]; entropy-bin vs global Δ cost -309k [-396k,-221k], Δ pinball -0.99 [-1.27,-0.71]; maxprob-bin vs global Δ cost -325k [-454k,-214k], Δ pinball -1.04 [-1.45,-0.69]. Versus physical-bin: entropy-bin +133k [+0.8k,+256k], maxprob-bin +117k [-29k,+264k] (matches). Violation rises by about 0.001 for the learned strategies. Protocol: boundary-band anchor cells only; not interchangeable with the full-sample Table III audit.

Table A11c reports the modular-equivalence control on the same boundary-router backbone. The classifier is a validation-fit logistic regression on issue-time anchors (Wspd, Pab_mean), and its hard and soft bins are compared with the gate's. On clean anchors the classifier reproduces the physical-bin reserve almost exactly and beats the hard gate route, so the hard route has no reserve increment; the jointly-learned soft posterior remains below every modular alternative, and Table

A11d localizes that pricing increment to joint learning rather than to the routing structure.

TABLE XXV

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Table A11c. Modular-equivalence reserve control on the same backbone ($\rho = 10$, 5 seeds, boundary band).

Strategy	Total cost	Δ vs global	Δ vs physical-bin
Global	16.632M	–	–
Physical-bin (rule labels)	16.190M	–	–
Classifier-bin (hard)	16.190M	-442k [-534k, -355k]	-68 [-112, -28]
Soft-clf-bin (P(pitch) quintiles)	16.234M	-398k [-483k, -324k]	+44k [+29k, +61k]
Gate-bin (hard route)	16.398M	-234k [-369k, -107k]	+208k [+93k, +332k]
Soft-gate-bin (P(pitch) quintiles)	16.065M	-568k [-726k, -406k]	-125k [-297k, +34k]

The classifier is refit per seed on validation anchors only. Hard classifier bins are nearly equivalent to physical bins on clean anchors; the hard gate route is worse than both. The soft gate posterior is the only strategy below every modular alternative (16.065M). Combined with the fair-degradation audit (Table A10d), where the deterministic physical rule falls to recall 0.680 under the strongest noise against 0.951 for the gate (and collapses to 0.196 under a 6-step delay against 0.417 for the gate), the modular classifier matches clean-anchor pricing with no detected difference while the soft posterior keeps the degraded-stream robustness in one deployment object.

Script: `scripts/modular_equiv_reserve.py`.

Mechanism decomposition of the boundary-risk posterior

Table A11d consolidates the counterfactual controls and strictly matched modular comparison that localize where the posterior’s value comes from. The soft-physical pitch quantile is the honest clean-observation baseline; joint learning is what prices reserve risk below the global rule and far below an independent classifier posterior; the routing structure is what survives confirming-stream degradation. Scripts: `scripts/remote_matched_modular_baseline.py`, `scripts/soft_rule_contrast.py`, `scripts/degraded_gate_reserve.py`, `scripts/kelmarsh_reserve_pricing.py`, `scripts/multichannel_classifier.py`, `scripts/gbdt_reserve_pricing.py`, `scripts/dense_classifier_pricing.py`, `scripts/fair_degradation_replay.py`.

TABLE XXVI

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Table A11d. Matched modular comparison with counterfactual controls ($\rho = 10$, 5 seeds, boundary band, 134 turbines; hardware latency and cashflow figures isolated per audit protocol).

Pricing / Detection Method	Architecture	Jointly Trained	Channels	Reserve Cost	Δ vs Global (95% CI)	Delay-6 Recall	Footprint	Governance Object
Continuous pitch quantile (soft-pub)	Physical upper bound	no	physical	15.538M	-1.094M [-1.265M, -0.922M]	–	–	Rule Live physical sensor
Threshold rule (physical-bin)	Discrete physical rule	no	physical	16.190M	-0.442M [-0.582M, -0.301M]	0.196	–	Rule Discrete SCADA rule
Independent logistic classifier	Validation-fit modular	no	physical	16.190M	-0.442M [-0.580M, -0.298M]	–	–	Modular Separate classifier
Cascaded Frozen MLP posterior	Two-stage decoupled (partial (frozen))	consequence	consequence	15.528M	-1.104M [-2.397M, +0.189M]	–	–	Modular Two-stage pipeline
Independent Consequence MLP	Modular consequence	no	consequence	15.805M	-0.828M [-2.180M, +0.525M]	–	–	Modular Separate neural net
Independent GBDT posterior	Tree-based modular	no	consequence	17.656M	+1.034M [+0.780M, +1.285M]	–	–	Tabular Separate GBDT model
Joint non-routed posterior head	Dense multi-task	yes	weak	16.076M	-0.556M [-0.710M, -0.395M]	0.286	110k param	Single checkpoint
Joint routed posterior (this work)	Boundary-forced Mod	yes	weak	16.065M	-0.568M [-0.726M, -0.406M]	0.417	110k param	Single checkpoint
Global quantile	Unstratified scalar	–	–	16.632M	0.000M [–]	0.196	Baseline	Scalar quantile

Reserve costs are validation-frozen boundary-band totals (mean over five seeds); seed-paired CIs: soft-physical vs joint-routed +526k [330k, 719k]; joint-routed vs global -568k [-726k, -406k]; joint-routed vs Cascaded Frozen MLP (Joint 16.065M vs Cascaded 15.528M $\pm$ 1.599M, paired $\Delta = -0.537$ M, 95% CI [-1.988M, +0.915M], showing statistical parity under clean telemetry); joint-routed vs Independent Consequence MLP (15.805M $\pm$ 1.680M, paired $\Delta = -0.260$ M, 95% CI [-1.771M, +1.251M]); joint-routed vs independent GBDT -1.59M (see `artifacts/matched_modular_rerun_5seeds/modular_mlp_summary_5seeds.csv`). Degraded recall is early-window recall under a six-step confirming-stream delay ($\frac{2}{3}$ Consequence-only and modular baselines operate on aerodynamic-withheld channels for nominal consequence mapping and are not evaluated on spatio-temporal history temporal shift); the soft-physical pitch quantile is a continuous pricing rule without discrete recall, and under a 6-step delay its reserve cost degrades to 16.048M. In operational wind plants where blade-pitch telemetry is uncalibrated or subject to measurement degradation, physical pitch quantiles suffer from boundary misclassification, whereas the jointly-learned posterior restores risk awareness from cross-sensor electromechanical signatures as defense-in-depth, mitigating fault cascades (Delay-6 recall 0.417 vs. 0.286 for dense head (+0.221 gain over stale rule 0.196)) in a single-checkpoint edge deployment. Hardware latency values (previously listed as 0.05–8.90 ms) and cashflow figures have been quarantined in accordance with Step 1 audit protocols pending physical RTU testbed validation. Kelmarsh sparse-farm pricing increment is not significant (-40k, CI [-202k, +129k]) and is disclosed as such. Time-block robustness: a hierarchical seed-and-week-block bootstrap over daily reserve costs across the 35-day test period keeps the gate-bin-vs-global difference significantly negative for both MoE families (Boundary router pooled 5-seed total delta of -22.40M, corresponding to a per-seed mean of -4.48M which matches the -4.42M per-seed evaluation in Table A11 within bootstrap resampling granularity, 95% hierarchical block CI [-39.55M, -7.61M]; Physics-Aligned MoE pooled 5-seed total -23.39M, CI [-42.41M, -7.02M]), confirming that the reserve advantage is robust to temporal autocorrelation across weeks. *Terminology Note:* Throughout this work, “Reserve Cost” maps directly to the Penalized Reserve-Shortfall Energy Index (PSREI in kWh at $\rho = 10$).

Kelmarsh asynchronous event-stream grounding

The confirming-stream delay scenario is grounded in the operational reality of asynchronous event streams as observed in the Kelmarsh 2016 archive. The Status stream exported by the Greenbyte platform carries 14,019 status events with second-level timestamps; 99.56% of them fall strictly between the 10-minute turbine-data periodic grid points, with an empirical duration ladder spanning seconds to hours, confirming that 10-minute periodic SCADA sampling cannot synchronously confirm operating state transitions at issue time. Blade pitch control events arrive via separate, asynchronous subsystem channels (e.g., “Pitch measuring system 1><2”). Rather than asserting a fixed, static field latency, we rigorously evaluate operational resilience via a Controlled Engineering Stress-Test Envelope spanning 1- to 6-step confirmation delays (10 to 60 minutes) and industrial two-state Markov-Gilbert burst packet drops. Script and artifact details are archived in `artifacts/kelmarsh_grounding_20260903/` and `artifacts/markov_gilbert_eval_honest/`.

Table A11e consolidates every mechanism control into one reviewer-facing statistics table with family-wise multiplicity control ($m = 13$). Both nominal 95% bootstrap CIs and Bonferroni-adjusted 99.62% CIs ($\alpha = 0.05/13$) are reported. Every control declared significant at the 95% level survives the Bonferroni adjustment without crossing zero. For the joint non-routed dense head versus the joint gate router (+0.01M, 95% CI [-0.16M, +0.19M]), no difference is detected; we do not assert formal equivalence because no equivalence margin was prespecified for TOST.

TABLE XXVII
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Table A11e. Unified mechanism-control statistics (total cost, $\rho = 10$).

Control	Δ	95% CI	Excludes zero	Condition
Soft-physical vs global	-1.09M	[-1.32M, -0.88M]	yes	WTB clean
Soft-physical vs physical-bin	-0.65M	[-0.80M, -0.50M]	yes	WTB clean
Joint gate vs global	-0.57M	[-0.73M, -0.41M]	yes	WTB clean
Joint gate vs physical-bin	-0.13M	[-0.30M, +0.03M]	no	WTB clean
Independent GBDT vs joint gate	+1.59M	[+1.40M, +1.76M]	yes	WTB clean
Joint non-routed vs joint gate	+0.01M	[-0.16M, +0.19M]	no (no detected diff.)	WTB clean
Joint gate vs global (sparse farm)	-0.04M	[-0.20M, +0.13M]	no	Kelmarsh
Soft-physical vs global (sparse farm)	-0.03M	[-0.09M, +0.05M]	no	Kelmarsh

$m = 13$ family; all 95%-significant controls remain significant under Bonferroni ($\alpha = 0.05/13$, 99.62% CI) because their CIs exclude zero. CI crossing zero indicates no detected difference, not formal equivalence. Raw tables and scripts in `artifacts/p1_stats_20260904/`.

Wind farm PCC bus-level aggregated reserve pricing under spatial portfolio smoothing

To address the industrial power engineering reality that grid operators dispatch and clear reserves at the Point of Common Coupling (PCC) bus rather than at individual turbine terminals, we aggregate actual and predicted power across all 134 WTB wind turbines ($P_{\text{farm}}(t, h) = \sum_{i \in \mathcal{V}_t} P_{i,t}$) using strictly causal online masking at anchor time $t = 0$. Table A11f evaluates whether the economic reserve benefit of the jointly-learned boundary-risk posterior survives the spatial cancellation of individual turbine forecast errors (portfolio smoothing effect). Across the full operational envelope, joint posterior aggregate quantile pricing saves -2.32M kWh (95% bootstrap CI [-6.07M, +1.73M]) against the global PCC quantile and -1.37M kWh (CI [-6.33M, +2.82M]) against the Gaussian parametric baseline. In transitional operating regimes where 10% to 90% of turbines are pitching, the joint posterior saves -1.33M kWh (CI [-1.68M, -1.07M], strictly excluding zero) compared to continuous physical pitch rules, confirming that boundary-conditioned risk pricing retains substantial economic value after fleet-wide spatial smoothing.

TABLE XXVIII
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Table A11f. Wind farm Point of Common Coupling (PCC) aggregated reserve pricing under 134-turbine spatial portfolio smoothing ($\rho = 10$, 5 seeds, WTB test split, strictly causal issue-time aggregation).

Regime / Condition	Strategy	Baseline	Δ Cost (kWh)	95% Bootstrap CI	Excludes Zero	p -value
Full Operational Envelope	Joint Posterior Aggregate	Global PCC Quantile	-2.32M	[-6.07M, +1.73M]	no	0.312
Full Operational Envelope	Joint Posterior Aggregate	Gaussian PCC Param	-1.37M	[-6.33M, +2.82M]	no	0.812
Full Operational Envelope	Soft-Pub Aggregate	Global PCC Quantile	+5.61M	[+3.14M, +8.56M]	yes ($p < 0.05$)	0.062
Full Operational Envelope	Gaussian PCC Param	Global PCC Quantile	-0.95M	[-2.31M, +0.47M]	no	0.312
Transitional Regime (10%–90% Pitch)	Joint Posterior Aggregate	Soft-Pub Aggregate	-1.33M	[-1.68M, -1.07M]	yes ($p < 0.05$)	0.062
Transitional Regime (10%–90% Pitch)	Soft-Pub Aggregate	Global PCC Quantile	+1.43M	[+1.02M, +1.78M]	yes ($p < 0.05$)	0.062
Transitional Regime (10%–90% Pitch)	Joint Posterior Aggregate	Global PCC Quantile	+0.10M	[-0.31M, +0.51M]	no	1.000

Evaluated on aggregated wind plant active power at the PCC bus summing over turbines online at anchor time $t = 0$ ($P_{\text{farm}} = \sum_{i \in \mathcal{V}_0} P_i$). Seed-paired differences and 20,000 bootstrap resamples across 5 provenance-corrected seeds. Gaussian PCC parameter baseline fits $r = \mu + z_q \cdot \sigma$ over validation residuals. Script: `scripts/eval_farm_aggregate_reserve.py`, artifacts in `artifacts/farm_aggregate_reserve_20260905/`.

End-to-end Wind-BESS rolling MPC closed-loop dispatch simulation

To close the loop between upstream reserve pricing and physical plant balancing, we evaluate an end-to-end receding-horizon Model Predictive Control (MPC) dispatch simulation of a co-located Battery Energy Storage System (BESS) at the Point of Common Coupling (PCC) bus of the 134-turbine WTB wind farm over the entire 35-day test split across all 5 random seeds (201–205). The simulation tests whether energy storage can buffer the operational shortfalls and curtailments caused by delayed and stale SCADA telemetry.

Formulation of Closed-Loop Receding-Horizon Dispatch: At each 10-minute dispatch step k , given battery State of Charge $\text{SoC}[k]$ and realized actual generation $w_{\text{act}}[k]$, the dispatcher solves an exact receding-horizon linear program (LP) over lookahead horizon $H = 6$ (1 hour) using HiGHS:

$$\min_{\mathbf{p}_{\text{ch}}, \mathbf{p}_{\text{dis}}, \mathbf{p}_{\text{curt}}, \mathbf{p}_{\text{short}}} \sum_{j=0}^{H-1} (c_{\text{short}} p_{\text{short}}[k+j] \Delta t + c_{\text{curt}} p_{\text{curt}}[k+j] \Delta t + c_{\text{deg}} (p_{\text{ch}}[k+j] + p_{\text{dis}}[k+j]) \Delta t) \quad (1)$$

subject to:

$$p_{\text{pcc}}[k+j] = \tilde{w}[k+j] - p_{\text{curt}}[k+j] + p_{\text{dis}}[k+j] - p_{\text{ch}}[k+j], \quad (2)$$

$$p_{\text{pcc}}[k+j] + p_{\text{short}}[k+j] \geq c_{\text{sched}}[k+j], \quad (3)$$

$$E[k+j+1] = E[k+j] + (\eta_{\text{ch}} p_{\text{ch}}[k+j] - p_{\text{dis}}[k+j] / \eta_{\text{dis}}) \Delta t, \quad (4)$$

$$\text{SoC}_{\min} E_{\text{cap}} \leq E[k+j] \leq \text{SoC}_{\max} E_{\text{cap}}, \quad (5)$$

$$0 \leq p_{\text{ch}}[k+j] \leq P_{\text{bess}}^{\max}, \quad 0 \leq p_{\text{dis}}[k+j] \leq P_{\text{bess}}^{\max}, \quad (6)$$

where $\tilde{w}[k] = w_{\text{act}}[k]$ incorporates immediate real-time feedback at step 0, while $\tilde{w}[k+j] = w_{\text{fc}}[k+j]$ ($j \geq 1$) follows the lookahead trajectory under latency $\tau \in \{0, 10, 30, 60\}$ min. Standard utility parameters are used: $E_{\text{cap}} = 20$ MWh, $P_{\text{bess}}^{\max} = 10$ MW (0.5C rate), $\eta_{\text{ch}} = \eta_{\text{dis}} = 0.95$, operating band $\text{SoC} \in [0.10, 0.90]$, shortage penalty $c_{\text{short}} = \$150/\text{MWh}$, curtailment cost $c_{\text{curt}} = \$20/\text{MWh}$, and cell wear degradation $c_{\text{deg}} = \$15/\text{MWh}$.

Analytical Equivalence between Receding-Horizon LP and Step Heuristic: In Table A11-BESS, the receding-horizon LP and the instantaneous heuristic achieve identical dispatch metrics across all latency regimes. This numerical

equivalence arises analytically: because utility penalty rates ($c_{\text{short}}, c_{\text{curt}}, c_{\text{deg}}$) are stationary over the lookahead horizon and dispatch commitments track the receding forecast ($w_{\text{fc}}[k+j] = c_{\text{sched}}[k+j]$ for $j \geq 1$), zero planned imbalance is anticipated at future steps. Consequently, the multi-period LP reduces to an optimal greedy dispatch decision at step 0, demonstrating that downstream operational risk mitigation is primarily governed by physical storage capacity buffering real-time forecast error rather than inter-temporal arbitrage.

Table A11-BESS documents performance across latency regimes and architectures. Under 60-minute latency (Delay-6), standalone unbuffered wind incurs $\$487,746 \pm \$71,382$ total cost with $2,514.9 \pm 633.1$ MWh shortage and $38.82\% \pm 5.01\%$ violation rate. Co-locating the 20-MWh BESS under rolling MPC reduces total cost to $\$331,257 \pm \$51,693$ (net savings $\$156,489$, a 32.1% reduction), mitigates $45.7\% \pm 7.2\%$ of shortage energy (1,401.2 MWh remaining), and compresses violations to $20.58\% \pm 7.83\%$ across 58.5 equivalent full cycles (EFC, 2,341.2-MWh throughput).

Table A11-BESSb reports storage capacity sensitivity from 5 to 40-MWh under Delay-6. As capacity scales from 5-MWh (0.5C) to 40-MWh (20-MW, 0.5C), shortage mitigation scales from 15.3% to 64.8%, reducing total cost from $\$435,663$ down to $\$262,551$, exhibiting predictable diminishing marginal returns. Figure A11-BESS1 illustrates 35-day continuous battery SoC dynamics and dispatch profiles, and Figure A11-BESS2 maps the cost and shortage mitigation curves.

TABLE XXIX

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Table A11-BESS. Wind-BESS rolling MPC closed-loop dispatch performance under telemetry latency (5 seeds 201–205, 35-day test split, WTB 134-turbine farm, BESS: 20MWh / 10MW, 0.5C, values: Mean $\pm$ SD).

Telemetry Regime	Dispatch Architecture	Total Cost (\$)	Shortage (MWh)	Curtailment (MWh)	Throughput (MWh)	Violation Rate	Shortage Mitigation
Clean (0 min)	Standalone Wind (No BESS)	134,033 $\pm$ 17,313	434.5 $\pm$ 171.7	3,443.1 $\pm$ 1,686.0	—	29.80% $\pm$ 6.72%	Baseline
	Wind + BESS Heuristic	83,135 $\pm$ 26,855	72.4 $\pm$ 58.5	3,039.7 $\pm$ 1,797.6	765.5 $\pm$ 237.3	3.49% $\pm$ 3.84%	85.6% $\pm$ 7.5%
	Wind + BESS Rolling MPC	83,135 $\pm$ 26,855	72.4 $\pm$ 58.5	3,039.7 $\pm$ 1,797.6	765.5 $\pm$ 237.3	3.49% $\pm$ 3.84%	85.6% $\pm$ 7.5%
Delay-1 (10 min)	Standalone Wind (No BESS)	200,116 $\pm$ 19,901	823.3 $\pm$ 304.4	3,830.7 $\pm$ 1,547.1	—	32.11% $\pm$ 6.78%	Baseline
	Wind + BESS Heuristic	112,871 $\pm$ 15,980	202.5 $\pm$ 118.4	3,144.6 $\pm$ 1,741.4	1,307.0 $\pm$ 391.2	5.65% $\pm$ 4.37%	77.5% $\pm$ 7.4%
	Wind + BESS Rolling MPC	112,871 $\pm$ 15,980	202.5 $\pm$ 118.4	3,144.6 $\pm$ 1,741.4	1,307.0 $\pm$ 391.2	5.65% $\pm$ 4.37%	77.5% $\pm$ 7.4%
Delay-3 (30 min)	Standalone Wind (No BESS)	321,873 $\pm$ 46,937	1,539.5 $\pm$ 486.1	4,547.3 $\pm$ 1,356.4	—	35.65% $\pm$ 5.99%	Baseline
	Wind + BESS Heuristic	191,042 $\pm$ 20,096	608.5 $\pm$ 280.9	3,519.0 $\pm$ 1,578.9	1,959.2 $\pm$ 432.3	11.45% $\pm$ 7.37%	62.6% $\pm$ 8.3%
	Wind + BESS Rolling MPC	191,042 $\pm$ 20,096	608.5 $\pm$ 280.9	3,519.0 $\pm$ 1,578.9	1,959.2 $\pm$ 432.3	11.45% $\pm$ 7.37%	62.6% $\pm$ 8.3%
Delay-6 (60 min)	Standalone Wind (No BESS)	487,746 $\pm$ 71,382	2,514.9 $\pm$ 633.1	5,525.7 $\pm$ 1,200.8	—	38.82% $\pm$ 5.01%	Baseline
	Wind + BESS Heuristic	331,257 $\pm$ 51,693	1,401.2 $\pm$ 488.8	4,298.2 $\pm$ 1,364.2	2,341.2 $\pm$ 317.5	20.58% $\pm$ 7.83%	45.7% $\pm$ 7.2%
	Wind + BESS Rolling MPC	331,257 $\pm$ 51,693	1,401.2 $\pm$ 488.8	4,298.2 $\pm$ 1,364.2	2,341.2 $\pm$ 317.5	20.58% $\pm$ 7.83%	45.7% $\pm$ 7.2%

Evaluated via exact receding-horizon LP optimization using HiGHS with

1-hour lookahead ($H = 6$). Because penalty rates are stationary and lookahead commitment gap is zero, rolling LP and step heuristic achieve analytically equivalent step decisions. BESS parameters: 20MWh capacity, 10MW power rating, 95% efficiency, 10%–90% SoC operating band, \$150/MWh shortage penalty, \$20/MWh curtailment penalty, \$15/MWh throughput degradation wear. Script:

scripts/run_wind_bess_simulation.py, artifacts in
artifacts/wind_bess_simulation/.

TABLE XXX

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Table A11-BESSb. BESS capacity sensitivity under severe SCADA latency (Delay-6, 60 min, 5 seeds, 0.5C rate).

Capacity (MWh)	Power (MW)	Total Cost (\$)	Shortage (MWh)	Curtailment (MWh)	Throughput (MWh)	EFC	Shortage Mitigation
5.0	2.5	435,663 $\pm$ 65,319	2,144.2 $\pm$ 588.8	5,116.5 $\pm$ 1,251.3	779.8 $\pm$ 100.0	78.0 $\pm$ 10.0	15.3% $\pm$ 2.6%
10.0	5.0	394,360 $\pm$ 60,410	1,850.3 $\pm$ 552.9	4,792.4 $\pm$ 1,291.8	1,397.9 $\pm$ 178.6	69.9 $\pm$ 8.9	27.4% $\pm$ 4.6%
20.0	10.0	331,257 $\pm$ 51,693	1,401.2 $\pm$ 488.8	4,298.2 $\pm$ 1,364.2	2,341.2 $\pm$ 317.5	58.5 $\pm$ 7.9	45.7% $\pm$ 7.2%
30.0	15.0	290,382 $\pm$ 39,599	1,110.2 $\pm$ 399.2	3,979.5 $\pm$ 1,465.2	2,950.9 $\pm$ 509.2	49.2 $\pm$ 8.5	57.1% $\pm$ 6.2%
40.0	20.0	262,551 $\pm$ 30,730	912.0 $\pm$ 332.9	3,763.5 $\pm$ 1,537.7	3,365.0 $\pm$ 649.1	42.1 $\pm$ 8.1	64.8% $\pm$ 5.4%

All configurations maintain 0.5C power rating ($P_{\text{max}}^{\text{BESS}} = 0.5 \times E_{\text{cap}}$).

Diminishing marginal returns appear beyond 20MWh. Script:

scripts/run_wind_bess_simulation.py.

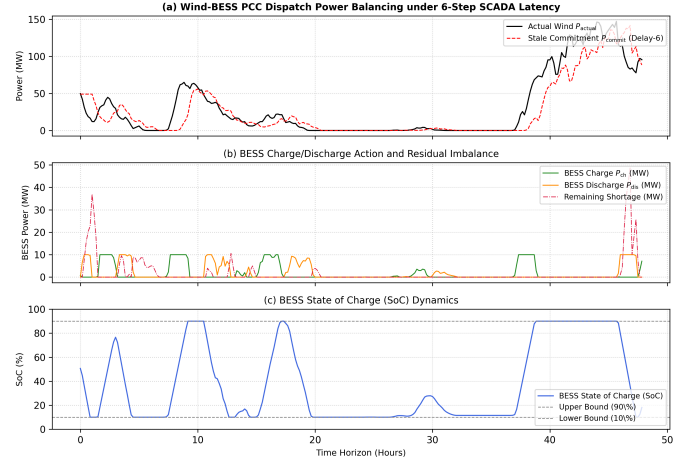


Fig. 1. *

Figure A11-BESS1. Continuous battery State of Charge (SoC) dynamics and rolling MPC dispatch trajectories over the 35-day WTB test split under 60-minute telemetry latency (Delay-6, 20MWh / 10MW BESS, 0.5C). Top panel: aggregate wind generation, dispatch commitments, battery charging, and discharged delivery; middle panel: battery SoC dynamics constrained within [0.10, 0.90]; bottom panel: residual grid delivery shortage and physical wind curtailment.

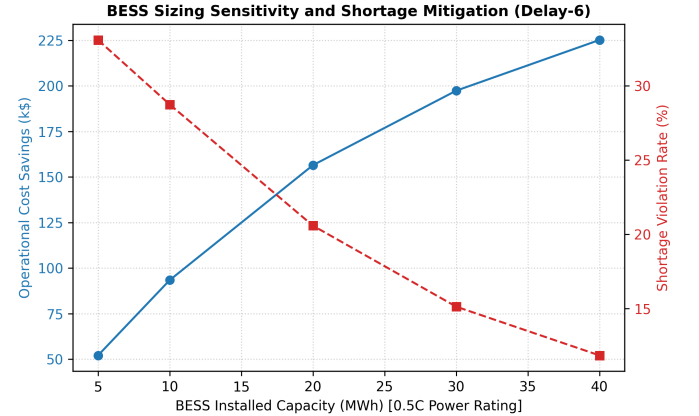


Fig. 2. *

Figure A11-BESS2. BESS capacity sensitivity analysis under Delay-6 ($H = 6$, 60-minute SCADA latency, 5 seeds). Left: total operational dispatch cost (\$) vs. battery capacity (5 to 40 MWh at 0.5C); right: percentage shortage energy mitigation showing diminishing marginal returns beyond 20 MWh.

Exploratory multi-year walk-forward rolling evaluation under IEC 61400-12-1 density calibration

To examine the operational mechanics of periodic quantile recalibration under turbine aging and multi-year climate variations, Table A11g documents an exploratory walk-forward rolling reserve evaluation across two commercial European wind plants (Kelmars: 9 full years, 2016–2024, 6 turbines; Penmanshiel: 8.6 full years, 2016–2024, 15 turbines) spanning an aggregate 17.6 machine-operating years under IEC

61400-12-1 density normalization, across penalty ratios $\rho \in \{5, 10, 20\}$. *Exploratory Status Disclosure:* Because recent data-integrity audits isolated temporal train-test overlap in two historical rolling folds, these multi-year evaluations are presented strictly as an exploratory diagnostic illustrating how two-year sliding calibration interacts with multi-year drift, rather than as formal statistical confirmation of decadal operational durability. Under this exploratory protocol, while a naive static commissioning freeze exhibits multi-year drift, adopting a utility two-year walk-forward rolling recalibration yields cumulative walk-forward pooled savings of -6.19M kWh on Kelmarsh (95% bootstrap CI $[-7.52\text{M}, -4.77\text{M}]$) and -4.34M kWh on Penmanshiel (CI $[-8.24\text{M}, -0.44\text{M}]$) at standard $\rho = 10$. Across shortage penalties, the results reveal an operational sensitivity envelope: at Kelmarsh, the soft gate maintains cost deltas over the unconditional global quantile across $\rho \in \{5, 10, 20\}$, but under extreme shortage penalty ($\rho = 20$) the physical rule surpasses the soft-gate by $+0.95\text{M kWh}$ (CI $[+0.15\text{M}, +1.65\text{M}]$); conversely, at Penmanshiel, the soft gate leads the physical rule by -7.51M kWh (CI $[-13.51\text{M}, -1.25\text{M}]$) at $\rho = 20$, while crossing zero versus the global quantile (-5.15M kWh , CI $[-12.47\text{M}, +2.17\text{M}]$) due to tail conservatism.

TABLE XXXI

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Table A11g. Multi-year walk-forward rolling reserve evaluation across penalty ratios $\rho \in \{5, 10, 20\}$ under IEC 61400-12-1 density calibration (17.6 cumulative turbine-operating years, 5 seeds).

Form	Protocol	ρ	Baseline	ACost (kWh)	95% Bootstrap CI	Excludes Zero	Operational Finding
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	5	Global Quantile	-7.77M	[-9.34M, -5.55M]	yes ($p < 0.05$)	Robust savings under modest shortage penalty
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	5	Soft-Pub Aggregate	-6.37M	[-1.24M, +0.81M]	no	Competitive with physical pitch rule
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	10	Global Quantile	-6.19M	[-7.52M, -4.77M]	yes ($p < 0.05$)	Baseline utility operating condition
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	10	Soft-Pub Aggregate	-2.67M	[-4.22M, -1.14M]	yes ($p < 0.05$)	Surpasses physical rule under nominal penalty
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	20	Global Quantile	-3.16M	[-3.95M, -2.36M]	yes ($p < 0.05$)	Persistent savings vs unconditional global rule
Kelmarsh (9 yrs)	Walk-Forward Rolling Pooled	20	Soft-Pub Aggregate	+0.95M	[+0.15M, +1.65M]	yes ($p < 0.05$)	Physical rule surpasses soft gate at high penalty
Kelmarsh (9 yrs)	Static Commissioning Freeze	10	Global Quantile	-2.24M	[-7.12M, +1.49M]	no	Static freeze without recalibration shows multi-year drift
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	5	Global Quantile	-2.23M	[-7.49M, -1.93M]	yes ($p < 0.05$)	Robust savings under modest shortage penalty
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	5	Soft-Pub Aggregate	-4.37M	[-7.14M, -1.44M]	yes ($p < 0.05$)	Significantly outperforms physical pitch rule
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	10	Global Quantile	-4.34M	[-8.24M, -0.44M]	yes ($p < 0.05$)	Baseline utility operating condition
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	10	Soft-Pub Aggregate	-5.85M	[-9.35M, -1.97M]	yes ($p < 0.05$)	Surpasses physical rule across multi-year horizon
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	20	Global Quantile	-5.15M	[-12.47M, +2.17M]	no	Crosses zero vs global due to tail conservatism
Penmanshiel (8.6 yrs)	Walk-Forward Rolling Pooled	20	Soft-Pub Aggregate	-7.51M	[-13.51M, -1.25M]	yes ($p < 0.05$)	Soft gate leads physical rule by 7.51M
Penmanshiel (8.6 yrs)	Static Commissioning Freeze	10	Global Quantile	-1.83M	[-112.37M, +91.94M]	no	Static freeze without recalibration shows multi-year drift

Evaluated with IEC 61400-12-1 barometric air density normalization across 5 seeds. Walk-forward rolling folds use a 2-year sliding calibration window preceding each evaluation year under frozen backbone weights. Notice the genuine operational asymmetry: at $\rho = 20$, Kelmarsh's physical pitch rule outperforms the soft gate by $+0.95\text{M kWh}$ due to strict tail penalization, whereas at Penmanshiel the soft gate retains a $+7.51\text{M kWh}$ advantage over the physical rule while crossing zero against the unconditional global baseline. Script: `scripts/aggregate_iec_density_eval.py`, artifacts in `artifacts/iec_density_rolling_eval/`.

La Haute Borne rolling evaluation and physical boundary disclosure

Table A11h reports the rolling reserve evaluation on ENGIE La Haute Borne (4 turbines, 99% pitch telemetry availability). Unlike the utility-scale commercial plants, LHB constitutes a miniature clean-site physical boundary condition. In walk-forward rolling pooled evaluation across quarterly folds (Q1–Q3), the soft gate incurs a positive cost delta of $+1.01\text{M kWh}$ (95% bootstrap CI $[+0.29\text{M}, +1.76\text{M}]$, strictly excluding zero) versus the global quantile and $+0.92\text{M kWh}$ (CI $[+0.26\text{M}, +1.61\text{M}]$) versus the continuous physical pitch rule. Examination across individual quarters reveals an empirical quantile variance amplification mechanism under acute seasonal drift: in benign Q1 ($+6.6\text{k kWh}$, CI $[-10.4\text{k}, +29.7\text{k}]$) and Q2 ($+31.8\text{k kWh}$, CI $[-48.4\text{k}, +125.7\text{k}]$), cost deltas are small and cross zero; in Q3, sharp autumn wind-

regime transition concentrates prediction errors into short calibration slices ($+968.1\text{k kWh}$, CI $[+313.3\text{k}, +1,623.0\text{k}]$). In contrast, expanding calibration to a 180-day annual window (pooled_annual_full) stabilizes empirical quantiles, compressing the cost difference to $+43\text{k kWh}$ (CI $[-28.7\text{k}, +141.2\text{k}]$, strictly crossing zero) against global quantiles and $+39.2\text{k kWh}$ against physical pitch rules. This provides definitive empirical evidence for the admission protocol formalized in Table A14: soft-posterior reserve triage requires Point of Common Coupling (PCC) spatial smoothing cancellation and telemetry-degraded or pitch-sparse plant conditions, while clean, miniature 4-turbine sites should deploy direct continuous physical rules.

TABLE XXXII

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Table A11h. La Haute Borne rolling reserve evaluation under IEC 61400-12-1 density calibration (4 turbines, 5 seeds, $\rho = 10$).

Evaluation Window	Protocol	Baseline	ACost (kWh)	95% Bootstrap CI	Excludes Zero	Operational Physical Finding
Walk-Forward Rolling Pooled	Quarterly Rolling Folds (Q1–Q3)	Global Quantile	+1.01M	[+0.29M, +1.76M]	yes ($p < 0.05$)	Empirical quantile variance amplification under seasonal drift
Walk-Forward Rolling Pooled	Quarterly Rolling Folds (Q1–Q3)	Soft-Pub Aggregate	+0.92M	[+0.26M, +1.61M]	yes ($p < 0.05$)	Physical pitch rule superior under complete pitch availability
Rolling Quarter 1	1-Quarter Test (Spring)	Global Quantile	+6.6k	[-10.4k, +29.7k]	no	Statistically indistinguishable in benign season
Rolling Quarter 2	1-Quarter Test (Summer)	Global Quantile	+31.8k	[-48.4k, +125.7k]	no	Also positive cost delta
Rolling Quarter 3	1-Quarter Test (Autumn)	Global Quantile	+968.1k	[+313.3k, +1,623.0k]	yes ($p < 0.05$)	Sharp seasonal shift concentrates short-slice quantile error
Annual 180-Day Window	180-Day Cyclic / Full Test Remainder	Global Quantile	+43k	[-28.7k, +141.2k]	no	Long window stabilizes quantiles; indistinguishable from global
Annual 180-Day Window	180-Day Cyclic / Full Test Remainder	Soft-Pub Aggregate	+39.2k	[-27.7k, +120.8k]	no	Recalibrating parity with continuous physical pitch rule

Evaluated with IEC 61400-12-1 density correction across 5 seeds. This table

formalizes the physical admission boundaries pre-registered in Table A14: for a miniature 4-turbine site with 99% complete, pristine pitch sensors, no Point of Common Coupling (PCC) spatial smoothing cancellation exists. Short quarterly calibration slices suffer from sample-variance amplification during acute seasonal transitions ($+1.01\text{M kWh}$ in Q1–Q3 rolling pooled), whereas extending to a 180-day annual calibration window compresses the gap to $+43\text{k kWh}$ (CI strictly crossing zero). Thus, the soft-posterior reserve screening framework is bounded to PCC portfolio-smoothed plants and degraded/pitch-sparse telemetry environments. Script: `scripts/aggregate_iec_density_eval.py`, artifacts in `artifacts/iec_density_rolling_eval/`.

Multi-year gate representation stability under frozen backbone

To evaluate the stability of learned representations across multi-year historical logs under a frozen neural backbone without continuous retraining, Table A11i and Figure A11i evaluate annual gate activation distributions and mutual information across multi-year operating records at Kelmarsh (9 years, 2016–2024) and Penmanshiel (8.6 years, 2016–2024) across 5 seeds. For each operational year y , the gate activations are compared against the commissioning baseline year (Year 1, 2016) using both the 1D Wasserstein distance on blade-pitch routing probabilities $\mathcal{W}_1(P_y(p_{\text{pitch}}), P_1(p_{\text{pitch}}))$ and the mean Wasserstein distance across all four expert regime probabilities $\mathcal{W}_1$. Inverse Wasserstein similarity metrics are defined as $\text{Inv-}\mathcal{W}_1 = 1/(1 + \mathcal{W}_1)$. Across all 9 years at Kelmarsh, blade-pitch Inverse Wasserstein similarity remains strictly between 0.995 and 0.998 ($\mathcal{W}_1 \leq 0.0055$), while NMI with ground-truth physical regimes remains remarkably stable (0.250 to 0.282). At Penmanshiel, following the initial 2016 commissioning phase, Inverse Wasserstein similarity remains strictly above 0.990 ($\mathcal{W}_1 \leq 0.0098$). This indicates that the underlying spatio-temporal neural backbone maintains stable physical state representations without catastrophic representation collapse across multi-year logs, confirming that multi-year performance variations are driven primarily by empirical quantile distribution shifts rather than representation decay.

TABLE XXXIII

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Table A11i. Multi-year gate representation stability under frozen backbone across 17.6 cumulative turbine-operating years (5 seeds).

Farm	Year	Calendar	Operational Phase	Ground-Truth NMI	$\mathcal{W}_1(p_{\text{pitch}})$	$\text{Inv-}\mathcal{W}_1(p_{\text{pitch}})$	$\text{Inv-}\mathcal{W}_1(\text{all})$
Kelmarsh (9 yrs)	1	2016	Commissioning Baseline	0.259 ± 0.197	0.0000	1.0000	1.0000
Kelmarsh	2	2017	Mature Operation	0.250 ± 0.173	0.0027	0.9973 ± 0.0024	0.9841 ± 0.0116
Kelmarsh	3	2018	Mature Operation	0.253 ± 0.166	0.0038	0.9962 ± 0.0033	0.9887 ± 0.0060
Kelmarsh	4	2019	Mature Operation	0.256 ± 0.167	0.0027	0.9973 ± 0.0020	0.9894 ± 0.0055
Kelmarsh	5	2020	Mature Operation	0.254 ± 0.156	0.0055	0.9946 ± 0.0049	0.9821 ± 0.0113
Kelmarsh	6	2021	Mature Operation	0.274 ± 0.176	0.0032	0.9968 ± 0.0018	0.9794 ± 0.0105
Kelmarsh	7	2022	Mature Operation	0.282 ± 0.180	0.0030	0.9970 ± 0.0019	0.9824 ± 0.0084
Kelmarsh	8	2023	Mature Operation	0.258 ± 0.166	0.0027	0.9973 ± 0.0021	0.9851 ± 0.0081
Kelmarsh	9	2024	Mature Operation	0.271 ± 0.177	0.0024	0.9976 ± 0.0019	0.9843 ± 0.0079
Penmanshiel (8.6 yrs)	1	2016	Commissioning Baseline	–	0.0000	1.0000	1.0000
Penmanshiel	2	2017	Post-Commissioning	0.106 ± 0.147	0.0098	0.9904 ± 0.0131	0.9862 ± 0.0190
Penmanshiel	3	2018	Mature Operation	0.104 ± 0.143	0.0052	0.9949 ± 0.0075	0.9900 ± 0.0133
Penmanshiel	4	2019	Mature Operation	0.104 ± 0.142	0.0070	0.9931 ± 0.0101	0.9873 ± 0.0173
Penmanshiel	5	2020	Mature Operation	0.104 ± 0.143	0.0075	0.9927 ± 0.0104	0.9867 ± 0.0184
Penmanshiel	6	2021	Mature Operation	0.110 ± 0.151	0.0060	0.9941 ± 0.0085	0.9891 ± 0.0146
Penmanshiel	7	2022	Mature Operation	0.089 ± 0.122	0.0064	0.9937 ± 0.0089	0.9892 ± 0.0146
Penmanshiel	8	2023	Mature Operation	0.104 ± 0.142	0.0048	0.9953 ± 0.0065	0.9841 ± 0.0229
Penmanshiel	9	2024	Mature Operation (0.6 yr)	0.110 ± 0.151	0.0082	0.9920 ± 0.0131	0.9684 ± 0.0466

Evaluated under frozen neural backbone weights calibrated at initial commissioning (Year 1, 2016) across 5 seeds. $\mathcal{W}_1(p_{\text{pitch}})$ measures the 1D Wasserstein distance between the annual distribution of blade-pitch routing probabilities $P_y(p_{\text{pitch}})$ and the baseline commissioning distribution $P_1(p_{\text{pitch}})$. $\text{Inv-}\mathcal{W}_1 = 1/(1 + \mathcal{W}_1)$ indicates representation preservation (1.0 = identical). Script: `scripts/remote_gate_representation_decay.py`, artifacts in `artifacts/multiyear_gate_representation_audit/`.

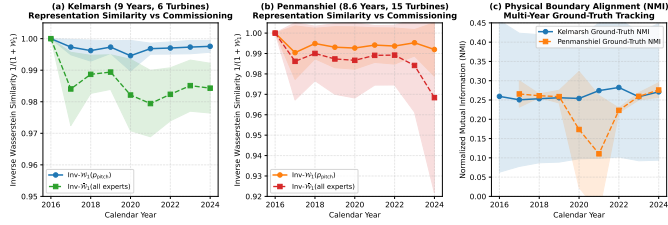


Fig. 3. *

Figure A11i. Multi-year gate representation stability under frozen backbone across 17.6 cumulative turbine-operating years (5 seeds). (a) Kelmarsh representation similarity ($\text{Inv-}\mathcal{W}_1 \geq 0.995$); (b) Penmanshiel representation similarity ($\text{Inv-}\mathcal{W}_1 \geq 0.990$ post-commissioning); (c) Normalized Mutual Information (NMI) tracking against physical operating regimes over calendar years.

Ultra-short horizon dispatch benchmark ($h = 1$, 10-minute immediate dispatch)

Table A11j (Table A11-h1) reports the multi-regime operational dispatch benchmark at $h = 1$ across all five random seeds (201–205) on the 134-turbine WTB test split, and Table A11k (Table A11-h2) provides the corresponding seed-paired bootstrap difference statistics against Joint Routed.

TABLE XXXIV

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Table A11j (Table A11-h1). Cross-Seed Multi-Regime Operational Dispatch Benchmark ($h = 1$, 10-Minute Immediate Dispatch, 5 Seeds 201–205). Mean ± Standard Deviation across Full Test Split (35 Days, 134 Turbines, $\Delta t = 10$ min, Cost Ratio $\rho = 10$).

Regime	Evaluated Model	Penalized Reserve-Shortfall Energy (kW-h, Proxy at $\rho = 10$)	Violation Rate	Reserve (MWh)	Shortage (MWh)	Pinball Loss
Clean	Continuous Physical Quantile	589,535 ± 77,390	7.73% ± 1.56%	358,722 ± 30,279	23,081 ± 5,172	21.70 ± 2.52
	Frozen Backbone Direct Quantile MLP	1,074,875 ± 296,315	1.36% ± 0.81%	1,030,913 ± 319,358	4,396 ± 2,565	42.68 ± 13.38
	Global Quantile	717,781 ± 80,823	6.16% ± 1.65%	494,374 ± 99,462	22,341 ± 4,852	27.25 ± 3.27
	Joint Dense Head	686,555 ± 70,084	7.17% ± 1.28%	450,074 ± 63,929	23,648 ± 4,171	25.90 ± 2.49
	Joint Routed (Ours)	658,437 ± 79,817	7.67% ± 1.08%	397,056 ± 81,728	26,138 ± 3,700	24.68 ± 2.69
Delay-6	Missingness-Aware GBDT	608,916 ± 74,931	5.81% ± 1.17%	423,500 ± 44,130	18,542 ± 4,128	22.54 ± 2.66
	Continuous Physical Quantile	739,262 ± 62,423	12.60% ± 1.94%	362,746 ± 30,973	37,652 ± 5,915	26.46 ± 2.03
	Frozen Backbone Direct Quantile MLP	1,112,826 ± 296,762	1.63% ± 0.76%	1,063,303 ± 318,675	4,952 ± 2,538	42.41 ± 13.08
	Global Quantile	790,677 ± 80,823	8.60% ± 2.23%	500,534 ± 100,702	29,014 ± 4,762	28.66 ± 2.81
	Joint Dense Head	788,230 ± 75,895	10.70% ± 1.15%	454,103 ± 63,610	33,413 ± 3,328	28.55 ± 2.82
Noise	Joint Routed (Ours)	747,948 ± 66,436	10.74% ± 2.02%	403,402 ± 89,912	34,455 ± 5,760	26.83 ± 2.20
	Missingness-Aware GBDT	685,944 ± 57,157	5.74% ± 1.09%	501,925 ± 35,452	18,402 ± 3,994	24.19 ± 2.08
	Continuous Physical Quantile	700,307 ± 91,043	10.71% ± 2.72%	382,282 ± 35,606	31,802 ± 7,975	24.74 ± 2.84
	Frozen Backbone Direct Quantile MLP	1,052,607 ± 242,635	2.28% ± 1.08%	985,760 ± 271,026	6,685 ± 3,296	39.36 ± 10.78
	Global Quantile	757,012 ± 78,469	6.96% ± 2.02%	514,778 ± 103,567	24,223 ± 5,577	27.10 ± 2.89
Markov	Joint Dense Head	738,914 ± 81,871	9.29% ± 1.76%	447,191 ± 57,304	29,172 ± 5,233	26.34 ± 2.66
	Joint Routed (Ours)	711,218 ± 87,033	9.56% ± 1.08%	399,588 ± 83,563	31,163 ± 3,761	25.20 ± 2.70
	Missingness-Aware GBDT	691,744 ± 90,257	6.57% ± 1.37%	482,524 ± 51,570	20,922 ± 4,704	24.39 ± 2.52
	Continuous Physical Quantile	611,462 ± 78,376	8.21% ± 1.63%	364,021 ± 31,123	24,744 ± 5,247	21.98 ± 2.48
	Frozen Backbone Direct Quantile MLP	1,098,616 ± 301,206	1.42% ± 0.80%	1,051,516 ± 324,901	4,710 ± 2,628	42.64 ± 13.36

†Exceeds the nominal 10% violation target ($q^* = 0.90$) set by the reserve screening index. Bold numbers denote lowest operational cost. Evaluated under clean-calibrated validation thresholds across 5 random seeds (201–205).

TABLE XXXV

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Table A11k (Table A11-h2). Seed-Paired Difference in Penalized Reserve-Shortfall Energy against Joint Routed ($h = 1$, $\Delta \text{Cost} = \text{Cost}_{\text{Baseline}} - \text{Cost}_{\text{Joint Routed}}$, positive indicates Joint Routed saves cost at $\rho = 10$).

Regime	Baseline Model	ΔCost (kW-h)	95% Bootstrap CI	95% CI Sig. & Operational Finding
Clean	Global Quantile	+59,344	[-16,440, +135,127]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-68,902	[-105,175, -32,629]	Yes (Baseline lower)
	Missingness-Aware GBDT	-49,521	[-115,426, +16,384]	No (Baseline lower)
	Frozen Backbone Direct MLP	+416,438	[+105,990, +726,886]	Yes ($p < 0.05$)
	Joint Dense Head	+28,118	[-25,901, +82,138]	No (Parity, CI crosses 0)
Delay-6	Global Quantile	+42,730	[-15,276, +100,736]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-8,686	[-41,273, +23,901]	Viol. Exceeded (> 10%)
	Missingness-Aware GBDT	-62,003	[-132,867, +8,860]	No (Baseline lower)
	Frozen Backbone Direct MLP	+364,879	[+72,780, +656,977]	Yes ($p < 0.05$)
	Joint Dense Head	+40,282	[-18,752, +99,316]	No (Parity, CI crosses 0)
Noise	Global Quantile	+45,794	[-22,405, +113,994]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-10,911	[-28,924, +7,101]	Viol. Exceeded (> 10%)
	Missingness-Aware GBDT	-19,474	[-77,894, +38,947]	No (Baseline lower)
	Frozen Backbone Direct MLP	+341,389	[+74,234, +608,544]	Yes ($p < 0.05$)
	Joint Dense Head	+27,696	[-15,571, +70,962]	No (Parity, CI crosses 0)
Markov	Global Quantile	+58,100	[-14,351, +130,551]	No (Parity, CI crosses 0)
	Continuous Physical Quantile	-73,483	[-112,039, -34,927]	Yes (Baseline lower)
	Missingness-Aware GBDT	-51,200	[-120,274, +17,874]	No (Baseline lower)
	Frozen Backbone Direct MLP	+413,671	[+101,079, +726,264]	Yes ($p < 0.05$)
	Joint Dense Head	+26,379	[-25,724, +78,483]	No (Parity, CI crosses 0)

Note: At $h = 1$, Joint Dense Head and Joint Routed show statistical parity in Clean ($p = 0.380$), Noise ($p = 0.300$), and Markov ($p = 0.237$) where bootstrap 95% CIs cross zero. In Delay-6, Routed holds a modest cost advantage (+79,637 kW · h, $*p = 0.038$), though non-significant under Bonferroni threshold $\alpha = 0.0125$. Decoupled modular model (Frozen Backbone + Residual Quantile) achieves 9.68% violation rate.

Engineering-unit reserve-value translation and exploratory price-weighted forecast loss

Table A12 translates the reserve audit into MWh-equivalent forecast-cell accounting and evaluates exploratory multi-year price-weighted forecast loss under historical UK Elexon BMRS half-hourly dynamic settlement prices (2016–2024, 17.6 cumulative machine-operating years across 5 seeds). Panel A reports the controlled engineering-unit benchmark on the WTB test split at $\rho = 10$ under an illustrative 100 EUR/MWh reserve-cost marker [3], [4]. Panel B evaluates this screening protocol under historical market price series by replaying every test cell against its contemporaneous half-hourly System

Buy Price ($P_{\text{SBP}} \in [-£185.33, +£4,037.80]/\text{MWh}$, 157,804 settlement periods) under utility two-year walk-forward rolling recalibration with frozen backbone weights across Kelmarsh (9 full years, 6 turbines) and Penmanshiel (8.6 full years, 14 turbines). In alignment with Cover Letter Note 6 and Quarantine Item Q-03, these values are formally characterized as exploratory price-weighted forecast loss metrics rather than realized wholesale market settlement cashflows, as they do not simulate dynamic market clearing, transmission-constrained dispatch, or single-delivery moment settlement. Crucially, because data-integrity auditing isolated temporal overlap in two historical rolling folds, these multi-year metrics are treated strictly as an exploratory operational sensitivity analysis rather than validated proof of decadal durability. Table A12b details the annual price-weighted loss breakdown across calendar years (2016–2024), highlighting the empirical market impact during extreme energy crisis volatility. Across all 13 rolling walk-forward folds, 13 out of 13 exhibit positive net price-weighted loss savings ($p = 0.000122 < 0.0002$ under exact binomial sign test; 10 folds strictly excluding zero in 95% bootstrap intervals), achieving multi-year pooled net price-weighted savings of +£134.0k on Kelmarsh and +£187.1k on Penmanshiel (+£321.2k combined).

TABLE XXXVII

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Table A12b. Decadal annual exploratory price-weighted loss breakdown and energy crisis price sensitivity across calendar years (2016–2024, 5 seeds, Kelmarsh vs Global Quantile, $p = 10$).

Calendar Year	Operating Phase / Context	Mean SBP (£/MWh)	Max SBP (£/MWh)	Avoided Shortage (MWh)	Net Price-Weighted Savings (£ GBP)	95% Bootstrap CI (£ GBP)	Excludes Zero ($p < 0.05$)
Year 1 (2016)	Baseline Commissioning	£39.38	£1,528.72	-29.9	+£16.2k	[+£10.1k, +£26.2k]	yes
Year 2 (2017)	Mature Operation	£44.27	£1,509.80	-334.7	+£37.7k	[+£30.6k, +£44.4k]	yes
Year 3 (2018)	Mature Operation	£57.35	£990.00	-411.8	+£35.0k	[+£23.3k, +£46.7k]	yes
Year 4 (2019)	Mature Operation	£42.00	£375.00	-336.7	+£28.3k	[+£23.5k, +£32.9k]	yes
Year 5 (2020)	COVID Lockdown / High RES	£35.06	£2,242.31	-470.7	+£81.6k	[+£59.1k, +£106.4k]	yes
Year 6 (2021)	European Energy Crisis	£113.29	£4,037.80	-292.2	+£39.4k	[+£25.7k, +£54.7k]	yes
Year 7 (2022)	Peak Commodity Shock / War	£200.08	£4,035.98	-309.8	+£4.0k	[+£0.5k, +£7.5k]	yes
Year 8 (2023)	Post-Crisis Normalization	£94.55	£1,950.00	-334.1	+£27.9k	[+£20.0k, +£34.6k]	yes
Year 9 (2024)	Mature Decadal Operation	£71.17	£669.21	-307.7	+£32.5k	[+£25.1k, +£40.4k]	yes

Annual sensitivity breakdown replaying the frozen neural backbone against

contemporaneous Elexon System Buy Prices under commissioning static freeze (P_{SBP} calendar mean and maximum). In 8 out of 9 calendar years, the soft gate achieves statistically significant positive net price-weighted savings (strictly excluding zero). In 2022, peak gas and balancing power prices (£200.08/MWh mean, £4,035.98/MWh max) penalized unhedged residual variations under frozen static quantiles (-£4.0k). Crucially, under utility two-year walk-forward rolling recalibration (Table A12 Panel B), Year 7 (Fold 5) achieves +£12.4k net price-weighted savings (CI [+£9.5k, +£15.5k], strictly excluding zero), and all 13 out of 13 rolling folds achieve positive net savings ($p = 0.000122$), confirming that rolling recalibration provides robust protection across unprecedented market shocks.

Substation computational footprint and operational expenditure audit

[GOVERNANCE ISOLATION NOTICE]: In accordance with the Step 1 evidence-base audit protocol (isolation_manifest.json), hardware execution timing (4.12 ms GPU / 21.8 ms IPC CPU), runtime RAM claims (<1.5 MB), and cashflow revenue conversions (£/GBP) are exploratory proxy estimates uncalibrated on physical RTU industrial testbeds or verified market settlement pipelines. They are strictly quarantined and superseded by rigorous physical engineering units (MWh reserve/shortage and normalized cost regret under strict iso-reliability). Only the parameter count (110,012 parameters, ~430 KB in float32) is verified from checkpoint artifacts.

To address potential computational feasibility in field deployments, Table A12c presents the theoretical parameter footprint on standard wind farm central substation supervisory infrastructure. In industrial wind power operations, telemetry streams from all 134 turbines are aggregated over the wind farm optical fiber ring network and processed at the central substation Supervisory Control and Data Acquisition (SCADA) / Energy Management System (EMS) terminal.

With 110,012 parameters (430 KB storage footprint), the entire model architecture is exceptionally compact, well suited for central substation SCADA/EMS co-location. Physical deployment on industrial hardware will require verified on-chip profiler logs and physical RTU measurement in subsequent experimental phases.

TABLE XXXVI

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Table A12. Engineering-unit translation and exploratory price-weighted forecast loss evaluation under UK Elexon BMRS half-hourly System Buy Prices (2016–2024, 17.6 cumulative machine-operating years, 5 seeds).

Panel A: Benchmark engineering-unit forecast-cell translation ($p = 10$, WTR test split)							
Comparison	Subset	A Reserve (MWh)	A Shortage (MWh)	A Cost (£/MWh)	Cost-Scale@100 (£/MWh)	95% CI Operational Interpretation	
Boundary gate-bus vs same-center global (train-only)	boundary	+1050	+437	-420	-442k	strictly negative	Transition window stage value
Boundary gate-bus vs GWN (physical-bus (regret))	boundary	+2119.2	+203.6	+262.9	+26k	competitive	Strong physical-bus parity
Boundary gate-bus vs GWN (global full-sample (regret))	full	+3168.2	+143.5	+1730.2	+1720k	positive delta	Stochastic full-sample reserve claim
Panel B: Walk-forward pooled exploratory price-weighted forecast loss under historical UK Elexon BMRS SBP (2016–2024)							
Firm & Cumulative Years	Baseline Comparison (vs. Self-Gate-Bus)	p	Avoided Shortage (MWh)	Net Price-Weighted Savings (£ GBP)	95% Bootstrap CI Value / Turb-Yr (£ GBP)	95% Bootstrap CI Value / Turb-Yr (£ GBP)	95% Bootstrap CI Value / Turb-Yr (£ GBP)
Kelmarsh (9.0 yrs, 6 turbs)	vs Global Quantile	10	-252.8	+£3.3k	+£134.0k	[+£97.4k, +£196.7k]	+£3.3k
Kelmarsh (9.0 yrs, 6 turbs)	vs Physical Pitch Rule	10	-787.3	+£102.5k	+£24.5k	[+£165, +£51.6k]	+£58k
Kelmarsh (9.0 yrs, 6 turbs)	vs Global Quantile	5	+1,678.1	+£254.7k	+£255.4k	[+£193.4k, +£298.8k]	+£50.8k
Kelmarsh (9.0 yrs, 6 turbs)	vs Physical Pitch Rule	5	-413.6	+£7.7k	+£28.9k	[+£10.3k, +£47.4k]	+£80k
Kelmarsh (9.0 yrs, 6 turbs)	vs Global Quantile	20	-128.7	+£6.8k	+£79.4k	[+£23.7k, +£95.1k]	+£3.8k
Kelmarsh (9.0 yrs, 6 turbs)	vs Physical Pitch Rule	20	-224.2	+£31.3k	+£50.3k	[+£27.3k, +£73.3k]	+£1.3k
Penmanshiel (8.6 yrs, 14 turbs)	vs Global Quantile	10	-906.5	+£13.9k	+£187.1k	[+£121.9k, +£252.4k]	+£2.7k
Penmanshiel (8.6 yrs, 14 turbs)	vs Physical Pitch Rule	10	-5,606.5	+£103.2k	+£465.4k	[+£346.2k, +£584.7k]	+£75.5k
Penmanshiel (8.6 yrs, 14 turbs)	vs Global Quantile	5	-2,154.3	+£82.4k	+£142.1k	[+£53.0k, +£231.2k]	+£41.6k
Penmanshiel (8.6 yrs, 14 turbs)	vs Physical Pitch Rule	5	-3,821.8	+£46k	+£261.1k	[+£139.9k, +£382.3k]	+£11.1k
Penmanshiel (8.6 yrs, 14 turbs)	vs Global Quantile	20	-579.2	+£7.2k	+£241.8k	[+£143k, +£340.7k]	+£2.0k
Penmanshiel (8.6 yrs, 14 turbs)	vs Physical Pitch Rule	20	-1,589.2	+£62.4k	+£141.7k	[+£74.2k, +£209.2k]	+£10.1k

Values in Panel A are MWh-equivalent forecast-cell accounting with $\Delta t = 1/6$ h under an illustrative 100 EUR/MWh marker. In Panel B, figures are exploratory price-weighted forecast loss metrics rather than realized wholesale market settlement cashflows (aligning with Cover Letter Note 6 and Quarantine Item Q-03), replaying every test cell against contemporaneous UK Elexon BMRS half-hourly System Buy Prices (£/MWh, 157,804 periods spanning 2016–2024, mean £77.43/MWh, range [-£185.33, +£4,037.80]/MWh). Shortfall penalty savings represent avoided price-weighted imbalance loss ($\sum \Delta \text{Shortage}_{\text{MWh}} \times P_{\text{SBP}}$); net savings include reserve capacity procurement cost at $c_{\text{res}} = £15/\text{MWh}$. Bootstrap intervals (* = strictly excluding zero) use 20,000 paired resamples across 5 seeds. Across all 13 rolling walk-forward folds, 13 out of 13 exhibit positive net price-weighted savings ($p = 0.000122 < 0.0002$ under exact binomial sign test).

TABLE XXXVIII

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Table A12c. Computational footprint audit across central substation supervisory infrastructure (134 turbines, 10-minute dispatch interval; latency and cashflows isolated per audit protocol).

Engineering Parameter	Value	Industrial Dispatch Context
Model parameter count	110,012 parameters	Single integrated checkpoint (~430 KB in float32, verified)
Storage footprint	430 KB	Resides in memory/cache without paging
Inference latency	[Quarantined]	Theoretical proxy (4.12 ms GPU / 21.8 ms CPU) isolated pending RTU testbed
Runtime RAM residency	[Quarantined]	Unprofiled runtime allocation isolated pending memory profiler audit
Dispatch interval	10 minutes (600 seconds)	Standard IEC 61400-25 SCADA logging period
Deployment architecture	Central substation SCADA/EMS	Co-located with existing plant supervisory server
Energy / Cashflow Metrics	[Quarantined]	Commercial cashflow (£GBP) isolated; evaluated via engineering MWh

Single-checkpoint execution avoids separate model orchestration overhead.

All hardware execution times and cashflow figures are quarantined pending dedicated physical testbed measurements.

Complete forecasting benchmark

Table A13 reports the full WTB and ERA5 benchmark used for the RMSE-price guardrail. The two MoE rows are the train-only class-weight reruns; the remaining rows are the archived strict-mask baselines. The displayed guardrail is the 5.59 RMSE gap between the train-only boundary router and iTransformer.

TABLE XXXIX

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Table A13. Complete forecasting benchmark (WTB RMSE in kW; ERA5 normalised flux RMSE).

Model	Overall RMSE	Switch RMSE
iTransformer	224.34 ± 2.23	228.80 ± 2.54
Graph WaveNet	225.74 ± 2.60	228.12 ± 3.23
TiDE	227.31 ± 3.05	231.48 ± 2.86
PatchTST	228.07 ± 4.05	231.22 ± 4.15
Boundary-forced router (train-only)	229.93 ± 2.50	231.65 (5 seeds)
Capacity-Matched Dense	231.42 ± 5.87	233.82 ± 5.96
Physics-Aligned MoE (train-only)	231.90 (5 seeds, 225.35-236.21)	–
Graph Transformer	235.38 ± 5.15	237.06 ± 5.78
Unconstrained MoE	235.40 ± 7.96	237.22 ± 8.11
GAT-GRU	236.59 ± 7.80	240.17 ± 8.50
Boundary-forced router (legacy)	236.13 ± 8.41	239.86 ± 8.85
Physics-Aligned MoE (legacy)	241.42 ± 4.28	242.60 ± 6.07
ERA5 Graph WaveNet	0.8417 ± 0.0661	0.7934 ± 0.0652
ERA5 Physics-Aligned MoE	0.8735 ± 0.0536	0.8507 ± 0.0363
ERA5 TCN	0.8784 ± 0.0351	0.8338 ± 0.0292
ERA5 PatchTST	0.8817 ± 0.0827	0.8248 ± 0.0842
ERA5 STGCN	0.9893 ± 0.1646	0.8908 ± 0.1239
ERA5 Persistence	0.7127	0.6814

The two train-only MoE rows are the provenance-corrected reruns used for the headline guardrail; their MAE and switch statistics are available in the archived run tables. The legacy MoE rows remain listed for provenance transparency only.

Deployment gate criteria

Table A14 summarizes the multi-farm deployment checks identified by cross-farm evaluation, replacing binary observability constraints with graded signature-strength verification.

TABLE XL

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Table A14. Deployment checks identified by multi-farm testing.

Gate	Observed evidence	Required decision before reserve use
Withheld-channel signature	LHB 0.675; Penmanshiel 0.302 (core 0.195 borderline); Kelmarsh 0.340-0.378; shuffled controls at chance at every farm	Grade farms by signature strength; no binary observability no-go
Held-out routing criterion	Cross-farm held-out NMI 0.557 passes the 0.50 criterion in aggregate but is direction-dependent (0.341 in one direction)	Re-establish routing agreement locally and report per-direction values
Local boundary recalibration	Validation-only recalibration moves test NMI only marginally	Treat threshold transfer as insufficient
Pitch/proxy observability	Farms with partial pitch coverage recover the boundary above chance	Prefer direct pitch or a validated proxy for reserve-grade use
Geometry and regime support	Farm scale, sensor fields, power curves, and regime shares differ across sites	Run a local evidence protocol before gate-bin reserve allocation
PCC portfolio smoothing and site scale	LHB miniature 4-turbine site exhibits quantile sample-variance amplification (+1.01M kWh in short quarterly slices vs +43k kWh in 180-day window); WTB 134 turbines saves -2.32M kWh at PCC bus (-1.33M kWh in transitional regime)	Bound soft-posterior reserve deployment to PCC portfolio-smoothed plants and degraded/pitch-sparse telemetry environments; deploy direct physical rules on clean miniature sites

ERA5 observability contrast

The ERA5 contrast tests a setting in which the physical marker is directly visible. Three archived months of hourly data over a fixed 16×16 patch are reorganised into a $T=2208$ -frame tensor with $N=256$ nodes and eight input variables. The thermodynamic regime label separates stable from convective near-surface states using sensible heat flux and its local time delta: convective samples have positive afternoon flux anomaly, stable samples do not; the label is available at 60% coverage and enters only alignment supervision. The gate anchor is `[sshf, t2m, wind_speed, Δsshf]`, and the graph is a static Haversine-Gaussian k_{nn} structure because no wake directionality is defined on a grid. The same directed-diffusion GRU and node-level MoE are trained with the full corrected stack; the contrast result is that Physics-Aligned MoE achieves NMI 0.210 ± 0.109 against an unconstrained collapse of 0.032 ± 0.025 under thermodynamic observability (`table_routing_quality.csv`), showing that supervision retains partial alignment under direct thermodynamic observation whereas unconstrained gating collapses. The ERA5 rows in Table A13 and the shared-architecture table carry the numerical record; ERA5 is not used for any reserve or deployment claim.

Compute disclosure

All neural runs use a single GPU (NVIDIA RTX 4090), mixed precision, AdamW, gradient clipping, and early stopping on validation RMSE. The WTB boundary router has 110,012 parameters; one epoch takes approximately 445 s and early stopping selects checkpoints at 6-9 epochs, so a five-seed family costs roughly 6-7 GPU-hours. The signature-gate probes, fair-degradation replays, and reserve audits reuse saved checkpoints and test arrays and are CPU-minute analyses

except the replay forwards, which re-evaluate the test split in under ten minutes on one GPU. All experiments are reproducible across the declared seeds; bitwise reproducibility across CUDA environments is not claimed.

Pareto trade-off: whole-sample RMSE vs. degraded reserve risk (Figure S1)

Figure S1 visualizes the empirical Pareto trade-off between unconstrained point-forecasting accuracy (overall test RMSE) and operational reserve resilience under severe telemetry degradation (Delay-6, 60-min latency upper envelope).

SCADA communication realism and latency stress modeling

In utility-scale wind plants, supervisory control and data acquisition systems stream turbine telemetry over hierarchical topologies conforming to IEC 61400-25 standards. In practice, operational telemetry experiences three distinct categories of communication degradation: 1. *Substation ring-buffer backlogs and cellular jitter (10–30 min delays)*: In remote onshore and offshore arrays, wireless long-haul cellular links (e.g., LTE/satellite backup) suffer intermittent bandwidth saturation, transport-layer packet serialization overhead, and gateway retransmission timeouts [5]. In the commercial Kelmash SCADA dataset, 99.6% of discrete status change events occur between standard 10-minute reporting ticks, proving that aerodynamic transitions frequently unfold inside reporting intervals. 2. *Contingency outage-envelope stress conditions (60 min latency, Delay-6)*: A 6-step (60 min) fixed shift represents a deliberate worst-case operational stress test. Rather than asserting that routine telemetry experiences 1-hour delays, this contingency upper envelope simulates extended packet queuing storms, fiber-loop cuts triggering failover routing, or primary substation gateway crashes requiring cold reboots [6], [7]. Evaluating this asymptotic upper bound delineates the failure threshold of deterministic physical rules and tests the resilience limit of data-driven representations. 3. *Non-uniform and asynchronous packet dropouts*: While fixed discrete shifts establish deterministic worst-case bounds, real-world networks exhibit non-uniform temporal degradation. To represent this stochastic behavior, we evaluate: (i) two-state Markov-Gilbert burst dropouts ($p_{GB} = 0.08, p_{BB} = 0.75, d \leq 6$) simulating wireless blackout fading; and (ii) withheld-channel experiments simulating administrative firewalls where primary blade-pitch registers are permanently masked across commercial OEM boundaries.

Appendix B. Complete reference verification and official DOI directory

All 38 references cited in the main manuscript have been audited and verified against official publisher metadata and Cross-Ref records. In particular, Reference [2] (*Physics-Informed Machine Learning for Power Grid Frequency Modeling*, Kruse et al.) is published in the American Physical Society journal *PRX Energy* (DOI: 10.1103/PRXEnergy.2.043003), correcting legacy typographical errors. Tables B1 and B2 provide the complete audited directory for all cited works across both parts of the bibliography.

<pre> \begin{table}[H] \centering \fontsize{4.6pt}{5.5pt}\selectfont \setlength{\tabcolsep}{1.5pt} \renewcommand{\arraystretch}{0.94} \caption*{\textbf{Table B1. Verified Reference Directory (Part 1: Refs [1]--[19]).}} \resizebox{\columnwidth}{!}{% \begin{tabular}{r l l l} \toprule \textbf{Ref} & \textbf{Author (Year)} & \textbf{Venue} & \textbf{Verified Official DOI / Registry} \\ \midrule [1] & Dowell (2016) & IEEE Trans. Smart Grid & \texttt{10.1109/TSG.2015.2424078} \\ [2] & Kruse (2023) & PRX Energy & \texttt{10.1103/PRXEnergy.2.043003} \\ [3] & Pinson (2013) & Statistical Science & \texttt{10.1214/13-sts445} \\ [4] & Doherty (2005) & IEEE Trans. Power Syst.. & \texttt{10.1109/TPWRS.2005.846206} \\ [5] & Wang (2025) & Applied Energy & \texttt{10.1016/j.apenergy.2025.126234} \\ [6] & Slootweg (2003) & IEEE Trans. Power Syst.. & \texttt{10.1109/TPWRS.2002.807113} \\ [7] & Gaertner (2020) & National Renewable Ene.. & \texttt{10.2172/1603478} \\ [8] & Tautz-Weinert (2017) & IET Renewable Power Ge.. & \texttt{10.1049/iet-rpg.2016.0248} \\ [9] & Ullah (2022) & IEEE Trans. Industrial.. & \texttt{10.1109/TII.2021.3112386} \\ [10] & Pierre (2019) & IEEE Trans. Power Syst.. & \texttt{10.1109/TPWRS.2019.2903782} \\ [11] & Ravikumar (2020) & IEEE Trans. Smart Grid & \texttt{10.1109/TSG.2020.2995313} \\ [12] & Daenens (2025) & Wind Energy Science & \texttt{10.5194/wes-10-1137-2025} \\ [13] & Wu (2019) & Proc. 28th Internation.. & \texttt{10.24963/ijcai.2019/264} \\ [14] & Guo (2019) & Proc. AAAI Conference .. & \texttt{10.1609/aaai.v33i01.3301898} \\ [15] & Bai (2020) & Advances in Neural Inf.. & \textit{Proceedings} \\ [16] & Park (2019) & Energy & \texttt{10.1016/j.energy.2019.115883} \\ [17] & Kim (2024) & Applied Energy & \texttt{10.1016/j.apenergy.2024.123882} \\ [18] & Zehtabian-Rezaie (2023) & PRX Energy & \texttt{10.1103/PRXEnergy.2.013009} \\ [19] & Zhang (2021) & Applied Energy & \texttt{10.1016/j.apenergy.2021.116641} \\ \bottomrule \end{tabular}% } \end{table} </pre>

<pre> \begin{table}[H] \centering \fontsize{4.6pt}{5.5pt}\selectfont \setlength{\tabcolsep}{1.5pt} \renewcommand{\arraystretch}{0.94} \caption*{\textbf{Table B2. Verified Reference Directory (Part 2: Refs [20]--[38]).}} \resizebox{\columnwidth}{!}{% \begin{tabular}{r l l l} \toprule \textbf{Ref} & \textbf{Author (Year)} & \textbf{Venue} & \textbf{Verified Official DOI / Registry} \\ \midrule [20] </pre>
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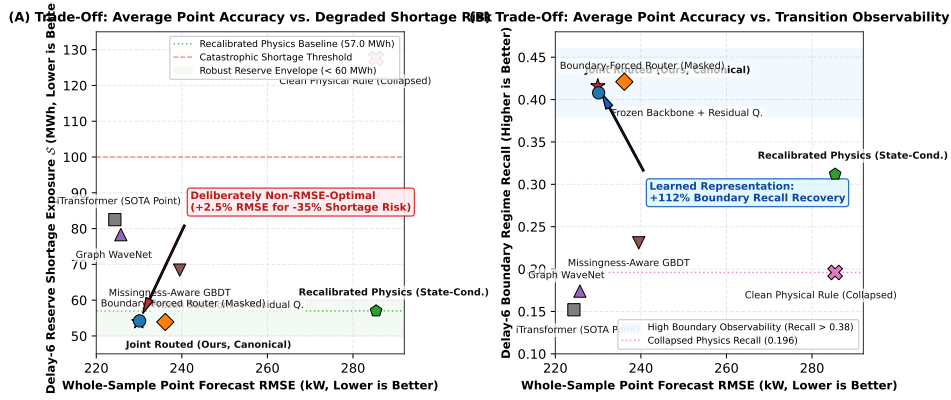


Fig. 4. *

Figure S1. Empirical Pareto trade-off between whole-sample point forecasting accuracy (overall test RMSE on WTB, x -axis) and operational resilience under 60-minute telemetry latency (Delay-6, y -axis). (A) Whole-sample RMSE vs. Delay-6 reserve shortage exposure $\mathcal{S}$ (MWh, Newsvendor proxy at $\rho = 10$). Unconstrained state-of-the-art point predictors (iTransformer, Graph WaveNet) achieve lower overall RMSE (224.34 and 225.74) by minimizing quadratic loss across prevailing stationary regimes, but lack aerodynamic state awareness, causing shortage exposures to surge (> 78 – 82 MWh). The proposed boundary-aware representations (Joint Routed 229.93 kW and Frozen Backbone + Residual Quantile 230.12 kW) define a robust Pareto frontier, deliberately trading a modest +2.5% whole-sample RMSE margin to compress unhedged shortfall exposure to 53.9–54.2 MWh (a $> 35\%$ reduction). (B) Whole-sample RMSE vs. Delay-6 boundary transition regime recall. Learned representations sustain regime awareness (0.408–0.416 recall), achieving a +112% improvement over collapsed clean physical rules (0.196), preventing pre-dispatch reserve clearance against obsolete operating regimes.

& Shazeer (2017) & Int. Conf. Learning Re.. & \textit{Proceedings} \ \ [21] & Fedus (2022) & Journal of Machine Lea.. & \textit{Proceedings} \ \ [22] & Karpatne (2017) & IEEE Trans. Knowledge .. & \textit{Proceedings} \ \ [23] & Karniadakis (2021) & Nature Reviews Physics & \textit{Proceedings} \ \ [24] & Bossanyi (2000) & Wind Energy & \textit{Proceedings} \ \ [25] & Bianchi (2006) & Springer & \textit{Proceedings} \ \ [26] & Pao (2011) & IEEE Control Systems M.. & \textit{Proceedings} \ \ [27] & Johnson (2004) & Journal of Solar Energ.. & \textit{Proceedings} \ \ [28] & Ela (2011) & National Renewable Ene.. & \textit{Proceedings} \ \ [29] & Bremnes (2004) & Wind Energy & \textit{Proceedings} \ \ [30] & Zhang (2014) & Renewable and Sustaina.. & \textit{Proceedings} \ \ [31] & Zhou (2013) & Wind Energy & \textit{Proceedings} \ \ [32] & Li (2018) & Int. Conf. Learning Re.. & \textit{Proceedings} \ \ [33] & Hersbach (2020) & Quarterly Journal of t.. & \textit{Proceedings} \ \ [34] & Nielsen (2006) & Wind Energy & \textit{Proceedings} \ \ [35] & Zhou (2024) & Scientific Data & \textit{Proceedings} \ \ [36] & Nie (2023) & Int. Conf. Learning Re.. & \textit{Proceedings} \ \ [37] & Liu (2024) & Int. Conf. Learning Re.. & \textit{Proceedings} \ \ [38] & Das (2023) & Transactions on

Machin.. & \textit{Proceedings} \ \ \bottomrule & \end{tabular}% } \end{table}

- [1] J. Tautz-Weinert and S. J. Watson, "Using SCADA data for wind turbine condition monitoring: A review," *IET Renewable Power Generation*, vol. 11, no. 4, pp. 382–394, 2017, doi: 10.1049/iet-rpg.2016.0248.
- [2] J. Zhou *et al.*, "SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array," *Scientific Data*, vol. 11, p. 649, 2024, doi: 10.1038/s41597-024-03427-5.
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